

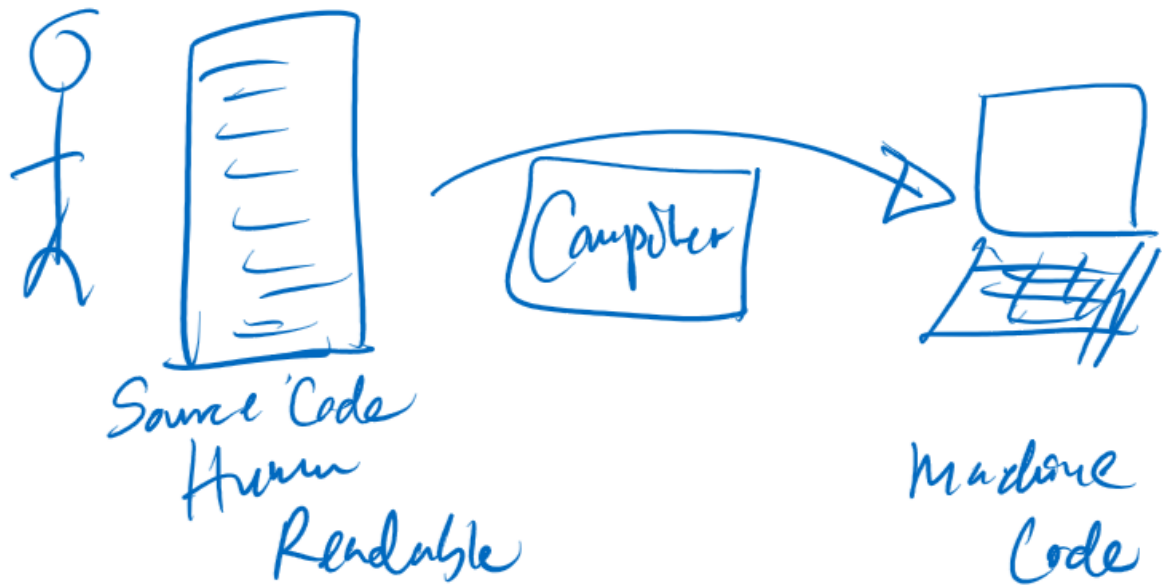
=====

[illegible]

////////////////////////////////////

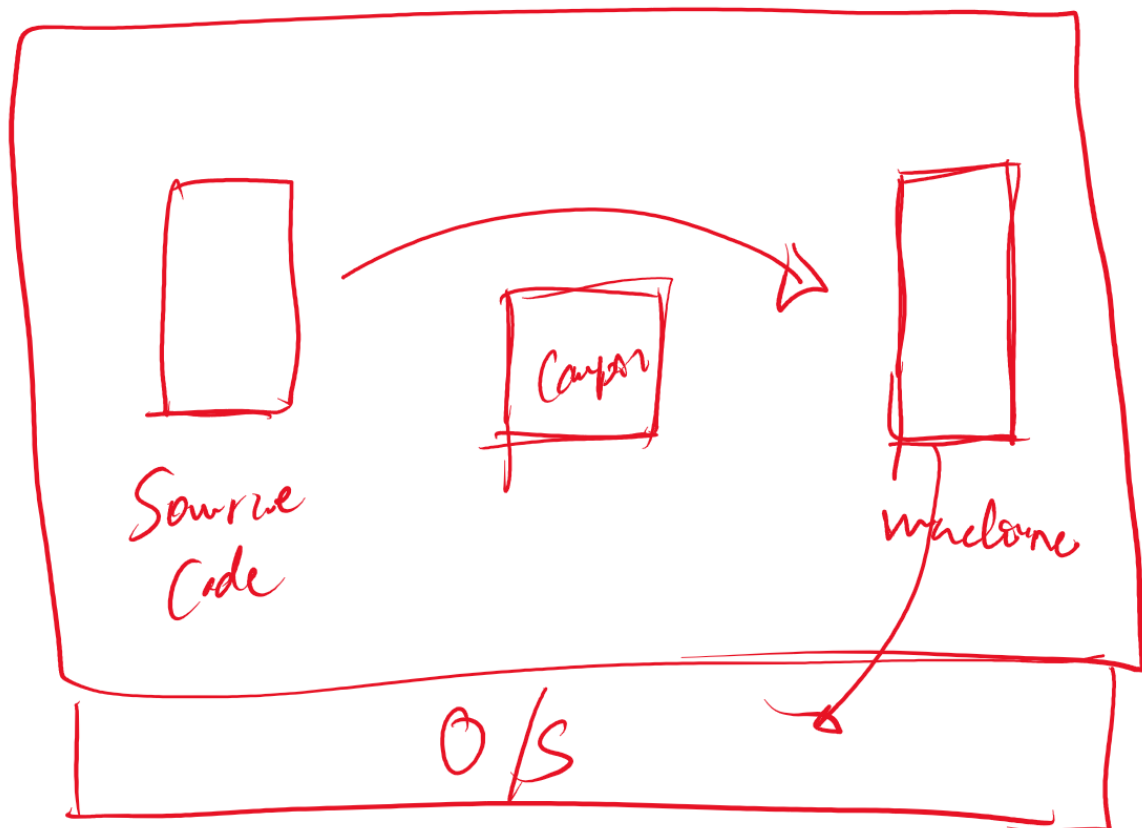


Q3



////////////////////////////////////

Q4



////////////////////////////////////

Q5

1992-3-4

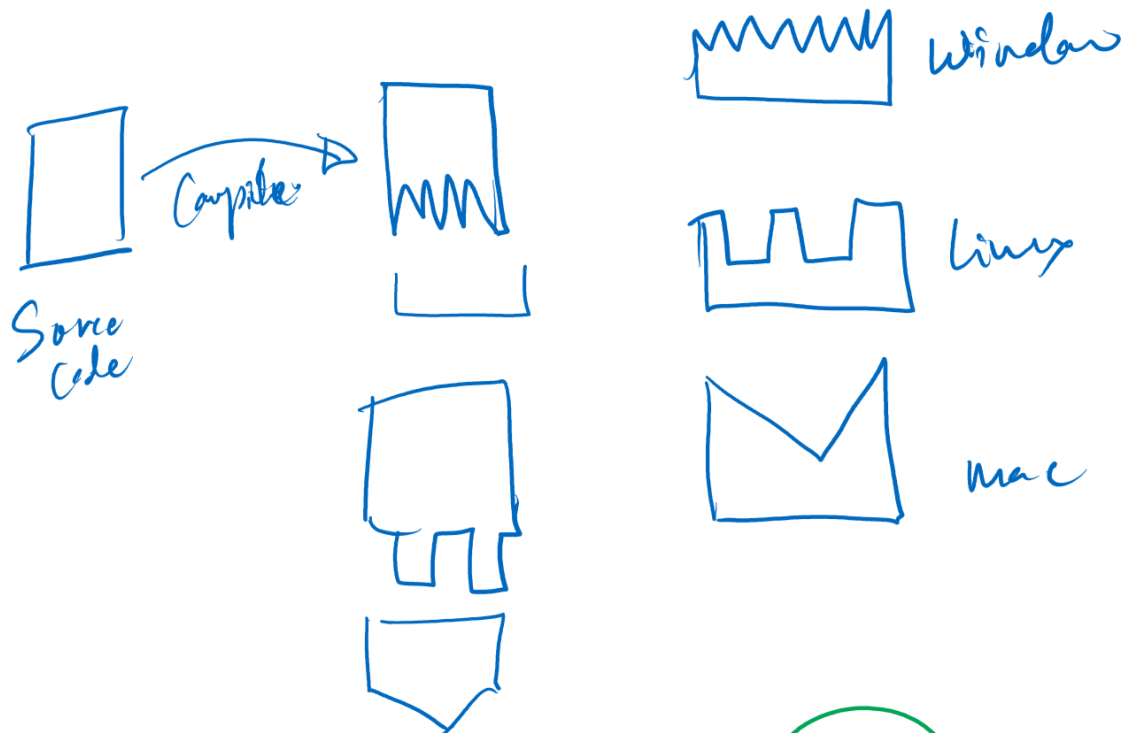
→ Java

Write Once, Run Anywhere

Platform Independence

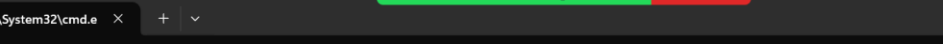
////////////////////////////////////

Q6



////////////////////////////////////

Q7



The screenshot shows a Windows Command Prompt window. The title bar at the top reads "C:\Windows\System32\cmd.e" and includes standard window controls. A green status bar at the very top of the window says "You are screen sharing" with a "Stop share" button. The command prompt displays the following text:

```
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

E:\ICM120_ICD121>notepad Example.java
```

////////////////////////////////////

Q9


```
    public static void main(String args[]){
        system.out.println("Hello World!");
    }
}
```

```
E:\ICM120_ICD121>javac Example.java
Example.java:3: error: package system does not exist
    system.out.println("Hello World!");
        ^
1 error

E:\ICM120_ICD121>
```

//

Q12

```
E:\ICM120_ICD121>java Example
Hello World!

E:\ICM120_ICD121>
```

//

Q13

Once we compile and create the bytecode, we no longer needed the source code to execute the program

//

Q14

```
class Example{
    public static void main(String args[]){
        System.out.println("A");
        System.out.println("B");
    }
}
```

```
System.out.println("C");
System.out.println("D");
System.out.println("E");
```

Q15\

```
class Example{
    public static void main(String
args[]){
        System.out.println("A");
        System.out.println("B");
        System.out.println("C");
        System.out.println("D");
        System.out.println("E");
    }
}
```

A	
B	
C	
D	
E	

Top to Bottom

E:\TCM120 TCD1215

Q16 main method - Java Entry point

```
File Edit View

class Example{
    public static void mai(String args[]){
        System.out.println("A");
        System.out.println("B");
        System.out.println("C");
        System.out.println("D");
        System.out.println("E");
    }
}

E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>javac Example.java

E:\ICM120_ICD121>java Example
Error: Main method not found in class Ex
ample, please define the main method as:
    public static void main(String[] args
)
or a JavaFX application class must exten
d javafx.application.Application

E:\ICM120_ICD121>
E:\ICM120_ICD121>
```

```
class Example{
    public static void mai(String args[]){
        System.out.println("A");
        System.out.println("B");
        System.out.println("C");
        System.out.println("D");
        System.out.println("E");
    }
}
```

////////////////////////////////////

class name

```
class Demo{
    public static void main(String[] args){
        System.out.println("Hello!");
    }
}
```

Name	Date modified	Type
Demo.class	3/1/2026 11:35 AM	CLASS File
Example.java	3/1/2026 11:32 AM	JAVA File

```
E:\ICM120_ICD121>
E:\ICM120_ICD121>javac Example.java

E:\ICM120_ICD121>java Demo
A
B
C
D
E
F

E:\ICM120_ICD121>
```

////////////////////////////////////

Q17 Legal Main Method

```
public static void main(String []args){
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q18

```
class Example{
    public static void main(String[] args){

    }
}
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q19

```
class Example{
    public static void main(String[] args){
        System.out.println("ABC");
        System.out.println();
        System.out.println("XYZ");
    }
}
```

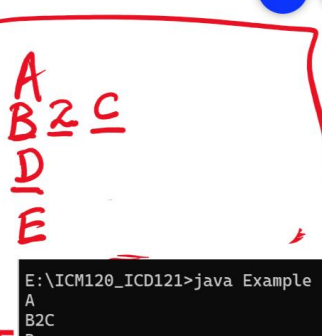
```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q20 - System.out.println();

```
class Example{
```


Q22

```
class Example{
    public static void main(String[] args){
        System.out.println("A");
        System.out.print("B");
        System.out.print("2");
        System.out.println("C");
        System.out.println("D");
        System.out.print("E");
    }
}
```



```
E:\ICM120_ICD121>java Example
A
B2C
D
E
E:\ICM120_ICD121>
```

Q23

Q24

```
class Example{
    public static void main(String[] args){
        System.out.print("A");
        System.out.print("B");
        System.out.print("");
        System.out.println("C");
        System.out.print(); // Error - Doesn't make any
sense
        System.out.print("D");
        System.out.println();
        System.out.print("E");
        System.out.print("E");
    }
}
```



```
}  
}
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Java Comments

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q25 - Line Comments

```
=====
```

```
class Example{  
    public static void main(String[] args){  
        System.out.println("A");  
        System.out.println("B");  
        //System.out.println("C");  
        System.out.println("D");  
        //System.out.println("E");  
        System.out.println("F");  
    }  
}
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q26

```
class Example{  
    public static void main(String[] args){  
        System.out.println("A");  
        System.out.println("B");  
  
        //System.out.println("C");System.out.println("D");  
        //System.out.println("E");  
        System.out.println("F");  
    }  
}
```



```
class Example{
    public static void main(String[] args){
        System.out.println("Hello!");
        System.out.println(10);
        System.out.println(22.34);
    }
}
```

Q30

Java Literals

```
class Demo{  
    public static void main(String[] args){  
        System.out.println("Hello!");  
        System.out.println(10);  
        System.out.println(22.34);  
    }  
}
```

directly $\rightarrow$ value

literals

Literals

- (1) Integer Literals
- (2) Floating Point Literals
- (3) Character Literals
- (4) String Literals
- (5) Boolean Literals
(true, false)
- (6) null Literals

Q31

```
class Example{
    public static void main(String[] args){
        String s = null;
        System.out.println("Hello!");
        System.out.println(10);
        System.out.println(22.34);
        System.out.println('A');
    }
}
```

////////////////////////////////////

```
class Example{
    public static void main(String[] args){
        System.out.println("Hello!"); // String Literals

        System.out.println(10); // Integer Literals

        System.out.println(22.345465); // Floating Point
Literals
        System.out.println(-22.345465); // Floating
Point Literals

        System.out.println('A'); // Character Literals

        System.out.println(true); // Boolean Literals
        System.out.println(false);

        String n = null; // null Literal
    }
}
```

////////////////////////////////////

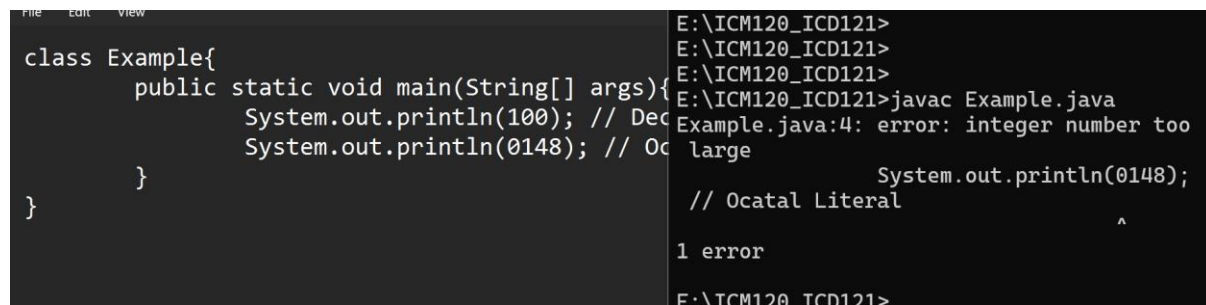
Q33 Integer Literals

```

=====
class Example{
    public static void main(String[] args){
        System.out.println(100); // Decimal Literal
        System.out.println(0144); // Ocatal Literal
    }
}

```

Q34



```

class Example{
    public static void main(String[] args){
        System.out.println(100); // Dec
        System.out.println(0148); // Oc
    }
}

```

```

E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>javac Example.java
Example.java:4: error: integer number too
large
        System.out.println(0148);
                        ^
1 error
E:\ICM120_ICD121>

```

Q35

```

class Example{
    public static void main(String[] args){
        System.out.println(100); // Decimal Literal (0-9)
        System.out.println(0144); // Ocatal Literal (0-7)
        System.out.println(0b10011001); // Binary
        System.out.println(0x64); // Hexa Decimal
        System.out.println(0x1A5F); // Hexa Decimal
        System.out.println(01); // Binary
        System.out.println(01); // Binary
    }
}

```

Q36

```

class Example{
    public static void main(String[] args){
        System.out.println(100); // Decimal Literal (0-9)
        System.out.println(0144); // Octal Literal (0-7)
        System.out.println(0b10011001); // Binary
        Literal (0-1)
        System.out.println(0x64); // Hexa Decimal
        Literal (0-15)
        System.out.println(0x1A5F); // Hexa Decimal
        Literal (0-9, A-F)

        System.out.println("0"+Integer.toOctalString(100));
    }
}

```

////////////////////////////////////

Q37

```

class Example{
    public static void main(String[] args){
        System.out.println(100); // Decimal Literal (0-9)
        System.out.println(0144); // Octal Literal (0-7)
        System.out.println(0b1100100); // Binary Literal
        (0-1)
        System.out.println(0x64); // Hexa Decimal
        Literal (0-15)
        System.out.println(0x1A5F); // Hexa Decimal
        Literal (0-9, A-F)

        System.out.println(Integer.toBinaryString(100));
    }
}

```

//

Q38 Floating Point Literals

```
class Example{  
    public static void main(String[] args){  
        System.out.println(100.0154);  
        System.out.println(1.000154E2);  
        System.out.println(10.00154E1);  
    }  
}
```

Handwritten notes illustrating floating point literals:

- 0, 1, 2, 3 - 9, 10, 11, 12, 13, 14, 15
0 - 9, A B C D E F
- 100.54
- 1.0054 × 10² (circled 2)
- 51.45
- 5.145 × 10
- 0.005145 × 10⁴ (circled 4)
- 0.005145F4
- 5.243E-3
- 0.005243
- 5.243 × 10⁻³ (circled)

//

Q39 Exercise

=====

```
class Example{  
    public static void main(String[] args){  
        System.out.println(100.0154);  
        System.out.println(1.000154E2);  
        System.out.println(10.00154E1);  
    }  
}
```

```

        System.out.println(0.005243); // Prints ->
0.005243
        System.out.println(); // Prints -> 0.005243
    }
}

```

//

Q40 From Q39

<pre> class Example{ public static void main(String[] args){ System.out.println(100.0154); System.out.println(1.000154E2); System.out.println(10.00154E1); System.out.println(0.005243); // System.out.println(5.243E-3); // } } </pre>	<pre> E:\ICM120_ICD121> E:\ICM120_ICD121>javac Example.java E:\ICM120_ICD121>java Example 100.0154 100.0154 100.0154 0.005243 0.005243 E:\ICM120_ICD121> E:\ICM120_ICD121> </pre>
---	--

//

Q41

<pre> class Example{ public static void main(String[] args){ System.out.println(100.0154); System.out.println(1.000154E2); System.out.println(10.00154E1); System.out.println(0.005243); // P System.out.println(5.243E-3); // P System.out.println(0.0000054123); } } </pre>	<pre> E:\ICM120_ICD121> E:\ICM120_ICD121>javac Example.java E:\ICM120_ICD121>java Example 100.0154 100.0154 100.0154 0.005243 0.005243 5.4123E-7 E:\ICM120_ICD121> E:\ICM120_ICD121> </pre>
---	--

//

Q42 Character Literals

=====


```

class Example{
    public static void main(String[] args){
        System.out.println('A');
        System.out.println('AB'); //Illegal
    }
}

```

//

Q43

```

class Example{
    public static void main(String[] args){
        System.out.println('A');
        System.out.println('B');
        System.out.println('\b'); // Backspace
        System.out.println('C');
        System.out.println('D');
    }
}

```

<pre> class Example{ public static void main(String[] args){ System.out.println('A'); System.out.println('B'); System.out.println('\b'); // Backspace System.out.println('C'); System.out.println('D'); } } </pre>	<pre> E:\ICM120_ICD121>javac Example.java E:\ICM120_ICD121>java Example A B C D E:\ICM120_ICD121> E:\ICM120_ICD121> E:\ICM120_ICD121> </pre>
--	---

//

Q44

```

class Example{
    public static void main(String[] args){
        System.out.println('A');
        System.out.print('B');
    }
}

```

```
System.out.println('\b'); // Backspace
```

```
System.out.println('C');
```

```
System.out.println('D');
```

```
}
```

```
}
```

```
class Example{
    public static void main(String[] ar
        System.out.println('A');
        System.out.print('B');
        System.out.println('\b'); //
        System.out.println('C');
        System.out.println('D');
    }
}
```

```
E:\ICM120_ICD121>javac Example.java
```

```
E:\ICM120_ICD121>java Example
```

```
A
```

```
B
```

```
C
```

```
D
```

```
E:\ICM120_ICD121>
```

```
E:\ICM120_ICD121>
```

```
E:\ICM120_ICD121>
```

```
E:\ICM120_ICD121>
```

//

Q45

```
class Example{
```

```
    public static void main(String[] args){
```

```
        System.out.println('A');
```

```
        System.out.print('B');
```

```
        System.out.print('\b'); // Backspace
```

```
        System.out.println('C');
```

```
        System.out.println('D');
```

```
    }
```

```
}
```

```
class Example{
    public static void main(String[] ar
        System.out.println('A');
        System.out.print('B');
        System.out.print('\b'); //
        System.out.println('C');
        System.out.println('D');
    }
}
```

```
E:\ICM120_ICD121>
```

```
E:\ICM120_ICD121>
```

```
E:\ICM120_ICD121>javac Example.java
```

```
E:\ICM120_ICD121>java Example
```

```
A
```

```
C
```

```
D
```

```
E:\ICM120_ICD121>
```

```
E:\ICM120_ICD121>
```

```
E:\ICM120_ICD121>
```

```
E:\ICM120_ICD121>
```

//

Q46

```
class Example{
    public static void main(String[] args){
        System.out.println('A');
        System.out.println('B');
        System.out.println('\b'); // Backspace
        System.out.println('C');
        System.out.println('D');
    }
}
```

A
B
C
D

A
B
C
D

```
class Example{
    public static void main(String[] args){
        System.out.println('A');
        System.out.print('B');
        System.out.println('\b'); // Backspace
        System.out.println('C');
        System.out.println('D');
    }
}
```

A
B
C
D

A
B
C
D

```
class Example{
    public static void main(String[] args){
        System.out.println('A');
        System.out.print('B');
        System.out.print('\b'); // Backspace
        System.out.println('C');
        System.out.println('D');
    }
}
```

A
B
C
D

////////////////////////////////////

Q47 = Escape Characters

=====

\b

\t

\n

\r

```

class Example{
    public static void main(String[] args){
        System.out.println("ABC\bEFG");
        System.out.println("ABC\tEFG");
        System.out.println("ABC\nEFG");
    }
}

```

```

E:\ICM120_ICD121>javac Example
ABEFG
ABC      EFG
ABC
EFG
E:\ICM120_ICD121>
E:\ICM120_ICD121>

```

//

Q48

```

class Example{
    public static void main(String[] args){
        System.out.println("\\");
    }
}

```

```

E:\ICM120_ICD121>javac Example
E:\ICM120_ICD121>java Example
\
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>

```

//

Q49

```

class Example{
    public static void main(String[] args){
        System.out.println("\\");    // print \
        System.out.println("\\\\");  // print \\
        System.out.println("/\\\\" ); // print /\
    }
}

```

```

class Example{
    public static void main(String[] args){
        System.out.println("\\");
        System.out.println("\\\\");
        System.out.println("/\\\\" );
    }
}

```

```

E:\ICM120_ICD121>
E:\ICM120_ICD121>javac Example
E:\ICM120_ICD121>java Example
\
\\
/\
E:\ICM120_ICD121>
E:\ICM120_ICD121>

```

//

Q50

```

class Example{
    public static void main(String[] args){
        System.out.println("\\");
    }
}

```

} }

```
E:\ICM120_ICD121>javac Example
E:\ICM120_ICD121>java Example
"
"
E:\ICM120_ICD121>
```

```
class Example{
    public static void main(String[] args){
        System.out.println(""); // Legal
        System.out.println(""); // Illegal
    }
}
```

```
class Example{
    public static void main(String[] args){
        System.out.println(""); // Legal
        System.out.println("\n");
    }
}
```

```
E:\ICM120_ICD121>javac Example
E:\ICM120_ICD121>java Example
'
'
E:\ICM120_ICD121>
```

////////////////////////////////////

```
class Example{
    public static void main(String[] args){
        System.out.println("Dinuka".length());
        System.out.println(162.length()); // illegal
    }
}
```

////////////////////////////////////

////////////////////////////////////

```
class Example{
```

```
public static void main(String[] args){
    int x;
    x = 100;
    System.out.println(x);
}
```

```
class Example{
    public static void main(String[] args){
        int x;
        x = 100;
        System.out.println(x);
    }
}
```

```
E:\ICM120_ICD121>javac Example.java
```

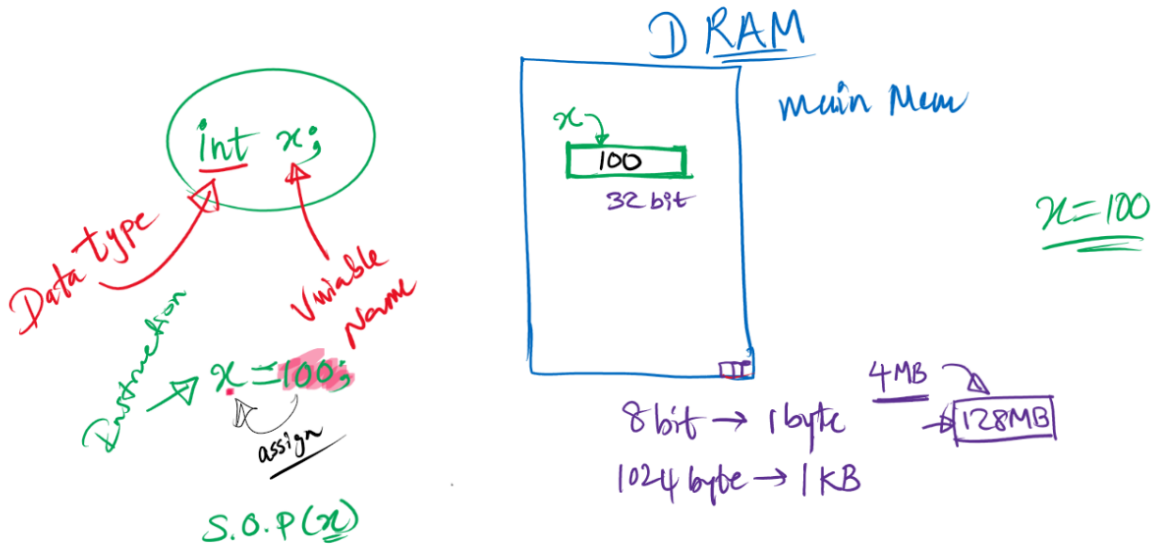
```
E:\ICM120_ICD121>java Example
100
```

```
E:\ICM120_ICD121>
```

```
E:\ICM120_ICD121>
```

```
E:\ICM120_ICD121>
```

```
E:\ICM120_ICD121>
```



////////////////////////////////////

Q56

```
class Example{
    public static void main(String[] args){
        int x;
        x = 100;
        x = 200;
        System.out.println(x);
    }
}
```

//

Q57 Compile Err

```
class Example{
    public static void main(String[] args){
        int x;

        System.out.println(x);
    }
}
```

<pre>class Example{ public static void main(String[] args){ int x; System.out.println(x); } }</pre>	<pre>200 E:\ICM120_ICD121>javac Example.java Example.java:5: error: variable x might not have been initialized System.out.println(x); ^ 1 error E:\ICM120_ICD121></pre>
--	---

//

Q58 Compile Err

```
class Example{
    public static void main(String[] args){
        int x;

        System.out.println(x);
        x = 100;
    }
}
```

//

Q59

```
class Example{
    public static void main(String[] args){
        int x;
```

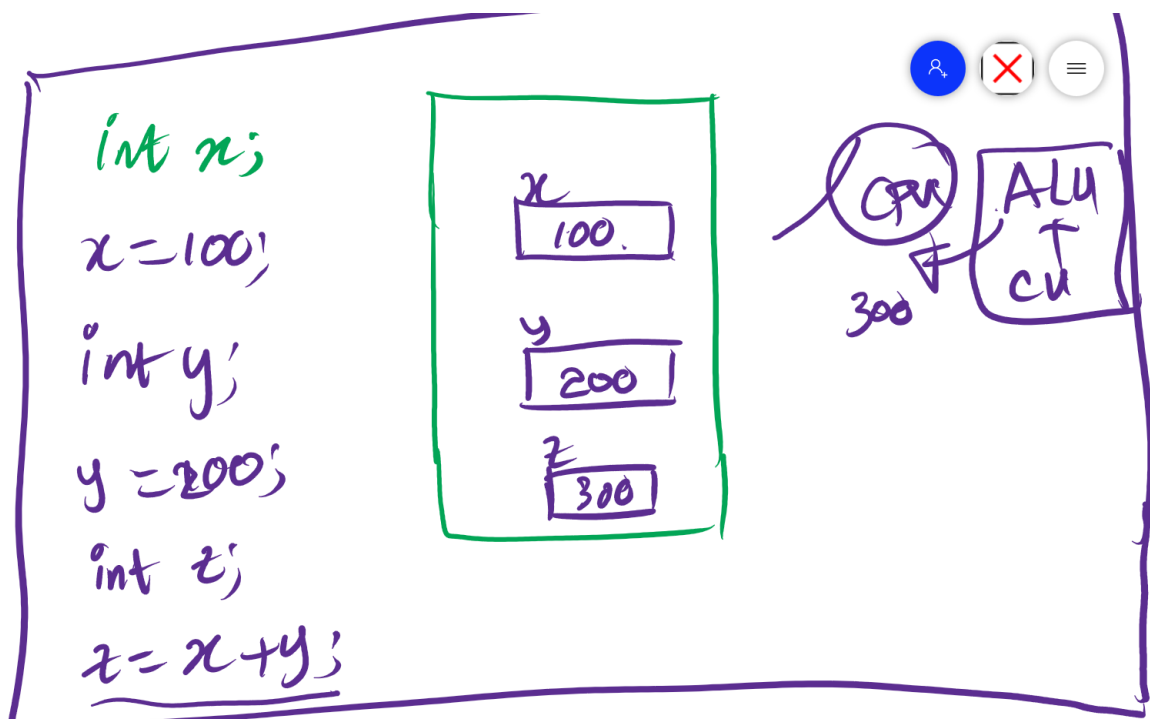

////////////////////////////////////

////////////////////////////////////

}

Q62

}



```

class Example{
    public static void main(String[] args){
        int x;
        x = 100;
        int y;
        y = 200;
        x = 500;
        int z;
        z = x + y;
        System.out.println(x);
        System.out.println(y);
        System.out.println(z);
    }
}

```

Variable Declaration

Variable Initialization

Assignment

//

Q63

```

class Example{
    public static void main(String[] args){
        int x = 100;

        int y, z;
        y = 200;

        z = x + y;
        System.out.println(x);
        System.out.println(y);
        System.out.println(z);
    }
}

```

//

Q64

```

class Example{

```

```

public static void main(String[] args){
    int x = 100, y, z;

    y = 200;

    z = x + y;
    System.out.println(x);
    System.out.println(y);
    System.out.println(z);
}
}

```

//

Q65

```

class Example{
    public static void main(String[] args){
        int x = 100, y, z;

        y = 200;

        int y; // Compile Error

        z = x + y;
        System.out.println(x);
        System.out.println(y);
        System.out.println(z);
    }
}

```

//

Q66

```

class Example{
    public static void main(String[] args){

```

```

        int x, y, z;

        x = y = z = 100;

        System.out.println(x);
        System.out.println(y);
        System.out.println(z);
    }
}

```

//

String Concatenation And Arithmetic Addition

//

Q67

```

class Example{
    public static void main(String[] args){
        System.out.println(10 + 20); // 30
        System.out.println("10" + "20"); // 1020
    }
}

```

//

Q68

```

class Example{
    public static void main(String[] args){
        System.out.println(10 + 20); // 30
        System.out.println("10" + "20"); // 1020
        System.out.println(10 + "20"); // 1020
    }
}

```

```
System.out.println("10" + 20); // 1020
```

```
    }  
}
```

<pre>class Example{ public static void main(String[] args){ System.out.println(10 + 20); // System.out.println("10" + "20"); System.out.println(10 + "20"); System.out.println("10" + 20); } }</pre>	<pre>E:\ICM120_ICD121>javac Example.java E:\ICM120_ICD121>java Example 30 1020 1020 1020 E:\ICM120_ICD121> E:\ICM120_ICD121></pre>
--	--

//

Q69

```
class Example{  
    public static void main(String[] args){  
        int x = 100, y = 200, z;  
  
        z = x + y;  
        System.out.println(); // 100 + 200 = 300;  
    }  
}
```

//

Q70

<pre>class Example{ public static void main(String[] args){ int x = 100, y = 200, z; z = x + y; System.out.println(""); // 100 + System.out.println(x+"ABC"); // System.out.println(10 + "ABC"); } }</pre>	<pre>E:\ICM120_ICD121> E:\ICM120_ICD121> E:\ICM120_ICD121> E:\ICM120_ICD121> E:\ICM120_ICD121> E:\ICM120_ICD121> E:\ICM120_ICD121>javac Example.java E:\ICM120_ICD121>java Example 100ABC 10ABC E:\ICM120_ICD121> E:\ICM120_ICD121> E:\ICM120_ICD121></pre>
--	--

//

Q71

```
class Example{
    public static void main(String[] args){
        int x = 101, y = 201, z;

        z = x + y;
        System.out.println(x + " + " + y + " = " + z); //
100 + 200 = 300

    }
}
```

//

Q72

```
import java.util.*;
class Example{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

        int x, y, z;

        x = input.nextInt();
        y = input.nextInt();
```

```
import java.util.*;
class Example{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

        int x, y, z;

        x = input.nextInt();
        y = input.nextInt();

        z = x + y;
        System.out.println(x + " + " + y + " = " + z);
    }
}
```

```

symbol:    variable input
location:  class Example
Example.java:7: error: cannot find symbol
    y = input.nextInt();
        ^
symbol:    variable input
location:  class Example
2 errors

E:\ICM120_ICD121>javac Example.java

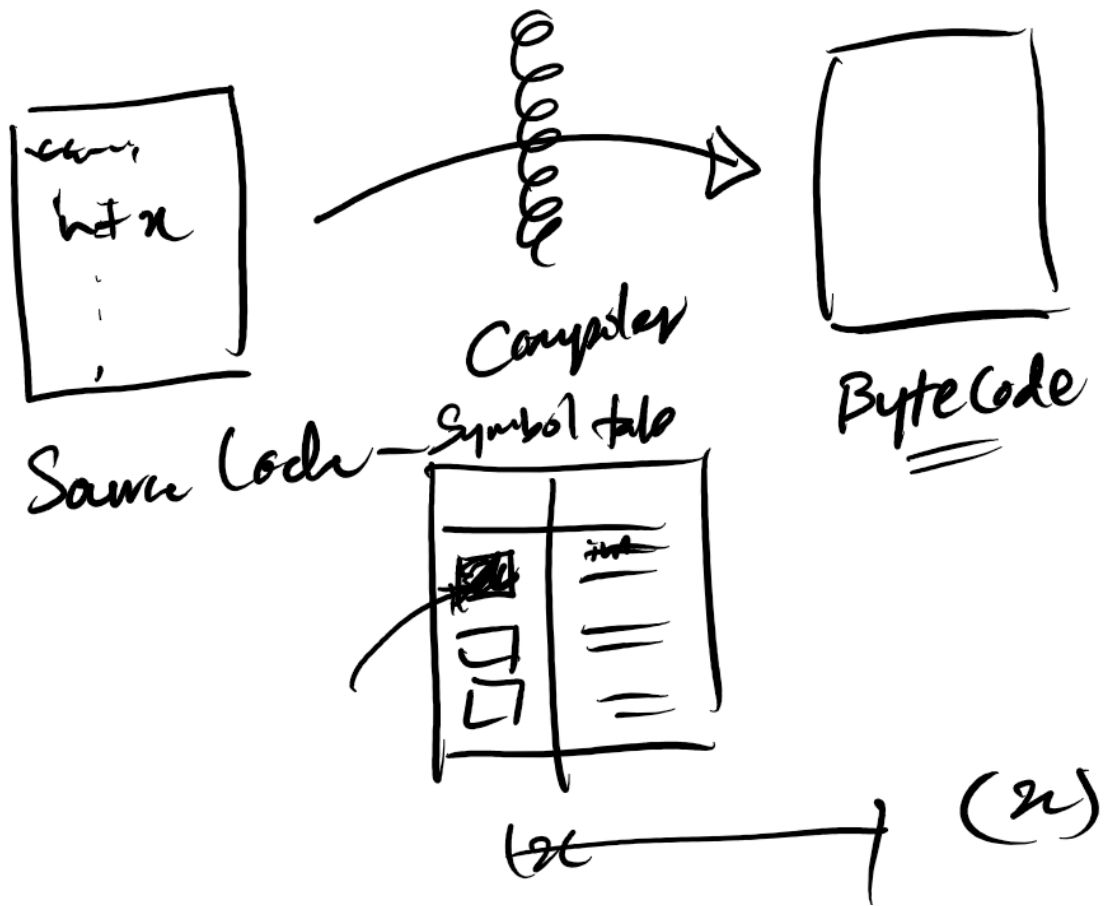
E:\ICM120_ICD121>java Example
10
20
10 + 20 = 30

E:\ICM120_ICD121>java Example
152
654
152 + 654 = 806

E:\ICM120_ICD121>java Example
85
65
85 + 65 = 150

E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
```

Q73

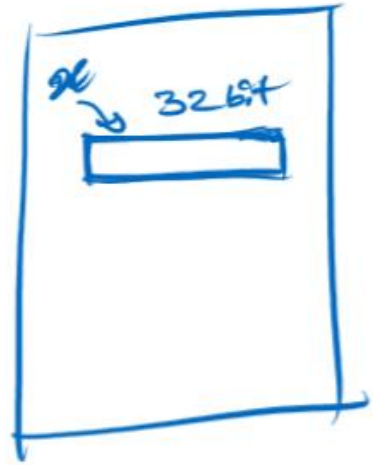


////////////////////////////////////

Primitive Data Types

////////////////////////////////////

int x;
↑
data type ↗
variable
name
(identified)



8 primitive data types

byte }
short } Integer
→ int } values
long }

float } Floating
double } Point values

char — characters
boolean — (logic)

byte 8 bits

Short 16 bits

int 32 bits

long 64 bits

float 32 bits

double 64 bits

Char 16 bits

boolean 1 bit

////////////////////////////////////

Q75

```
import java.util.*;
```

```
class Example{
```

```
public static void main(String[] args){
```

```
byte x;
```

```
x = 100;
```

```
System.out.println(x);
```

}

}

////////////////////////////////////

Q76

```
import java.util.*;
```

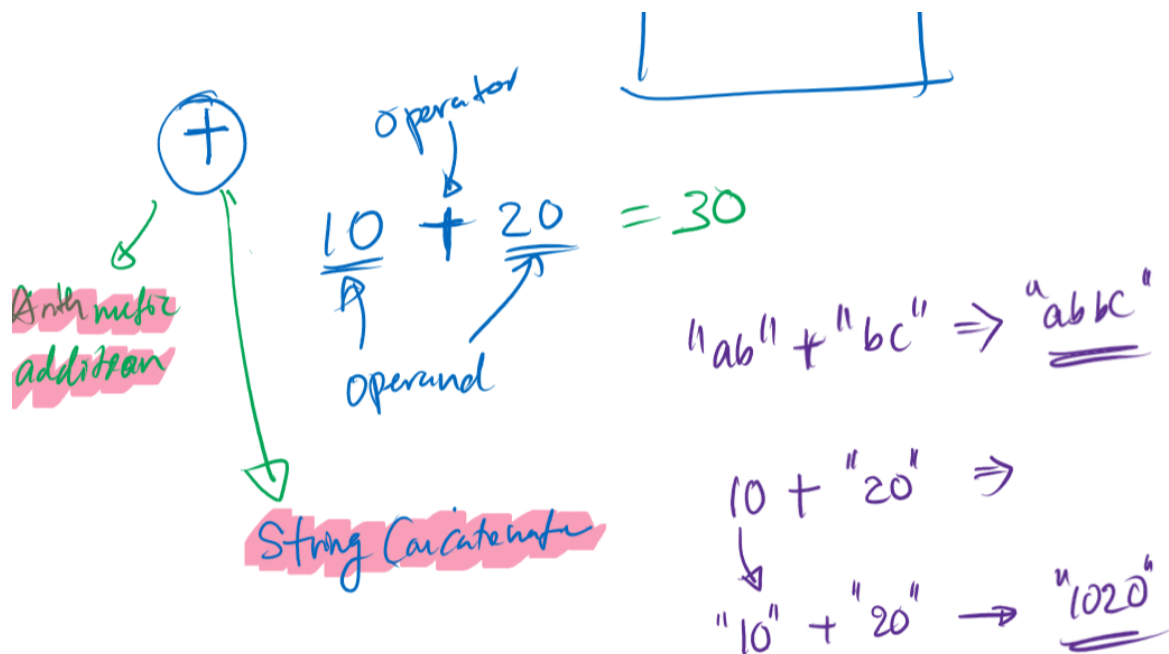
```

class Example{
    public static void main(String[] args){
        long y = 2154525852; // Compile Err.
        long x;
        x = 2154525852L; // Tells compiler to take this
as long Literal
        System.out.println(x);

        float f = 1.5f; // Tells compiler to take this as
float Literal
    }
}

```

////////////////////////////////////
Q77



////////////////////////////////////
Q78

```

import java.util.*;
class Example{
    public static void main(String []args){

```

```
System.out.println(10 + "20"); // 1020
```

```
System.out.println("A" + "B"); // AB
```

```
System.out.println('A' + 'B'); // 131
```

```
}
```

```
}
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q79

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String []args){
```

```
        int x = 100;
```

```
        int y = x;
```

```

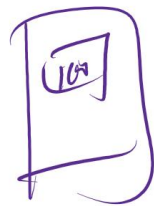
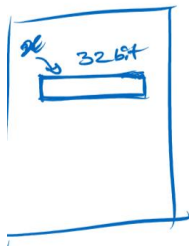
        System.out.println("x : " + x);
        System.out.println("y : " + y);
    }
}

```

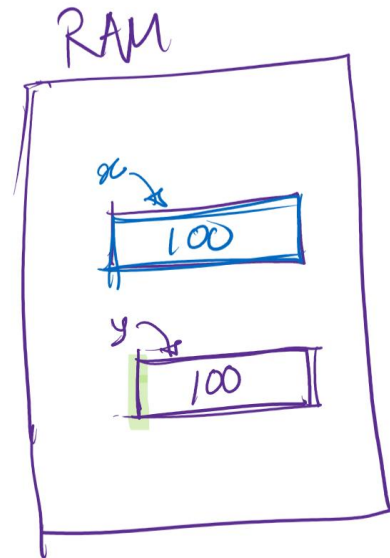
```

import java.util.*;
class Example{
    public static void main(String []args){
        int x = 100;
        int y = x;
        System.out.println("x : " + x);
        System.out.println("y : " + y);
    }
}

```



~~int x;~~
 32
~~int y;~~



////////////////////////////////////

Q80

```

import java.util.*;
class Example{
    public static void main(String []args){
        short s = 100;
        byte b = s; // Compile Error
    }
}

```

Hand-drawn diagram illustrating a memory layout. A large rectangle is divided into two horizontal sections. The top section is labeled "16 bit" and contains a smaller rectangle labeled "100". An arrow points from the letter "s" to the top-left corner of this rectangle. The bottom section is labeled "8 bit" and contains a smaller rectangle. An arrow points from the letter "b" to the top-left corner of this rectangle.

=====

////////////////////////////////////

=====

////////////////////////////////////

Q84 input double values


```
import java.util.*;
class Example{
    public static void main(String []args){
        Scanner input = new Scanner(System.in);
        double x;
        System.out.print("Input number: ");
        x = input.nextDouble();

        System.out.println("x : " + x);
    }
}
```

Q85 - I String variable

```
class Example{
    public static void main(String []args){
        String name;
        name = "dinuka";

        System.out.println("My Name is: " + name);
    }
}
```

Q86 String Input

```
import java.util.*;
class Example{
    public static void main(String []args){
```

```
E:\ICM120_ICD121>java Example
Enter your name: Dinuka
Your Name is: Dinuka

E:\ICM120_ICD121>java Example
Enter your name: Dinuka Kodithuwakku
Your Name is: Dinuka
```

```

////////////////////////////////////
Q87 input.nextLine();
=====
import java.util.*;
class Example{
    public static void main(String []args){
        Scanner input = new Scanner(System.in);
        String name;
        System.out.print("Enter your name: ");
        name = input.nextLine();
    }
}

```

```
E:\ICM120_ICD121>java Example
Enter your name: Dinuka Kodithuwakku
Your Name is: Dinuka Kodithuwakku
```

} }

```
E:\ICM120_ICD121>java Example
Enter your NIC: 564651656V
Enter your name: Your Name is:
```

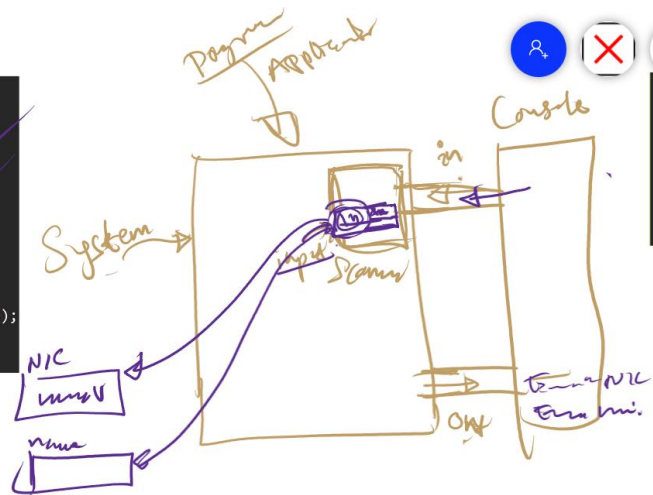
Q89

```
import java.util.*;
class Example{
    public static void main(String []args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your NIC: ");
        String nic = input.next();

        String name;
        System.out.print("Enter your name: ");
        name = input.nextLine();

        System.out.println("Your Name is: " + name);
    }
}
```



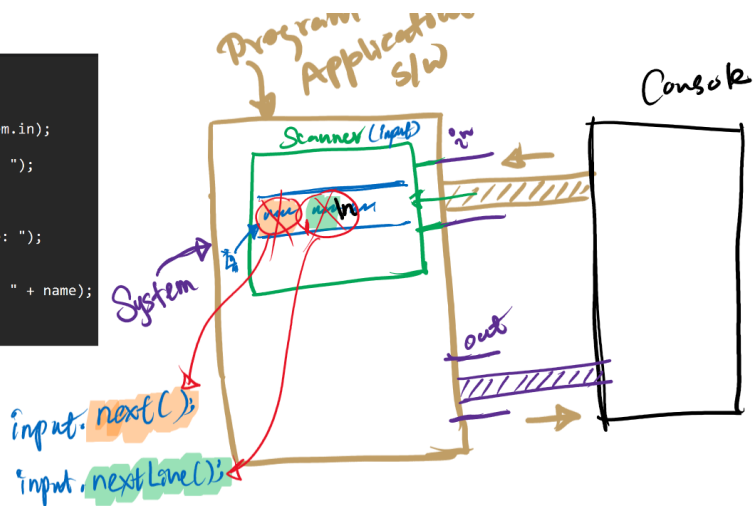
input.nextLine()

```
import java.util.*;
class Example{
    public static void main(String []args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your NIC: ");
        String nic = input.next();

        String name;
        System.out.print("Enter your name: ");
        name = input.nextLine();

        System.out.println("Your Name is: " + name);
    }
}
```

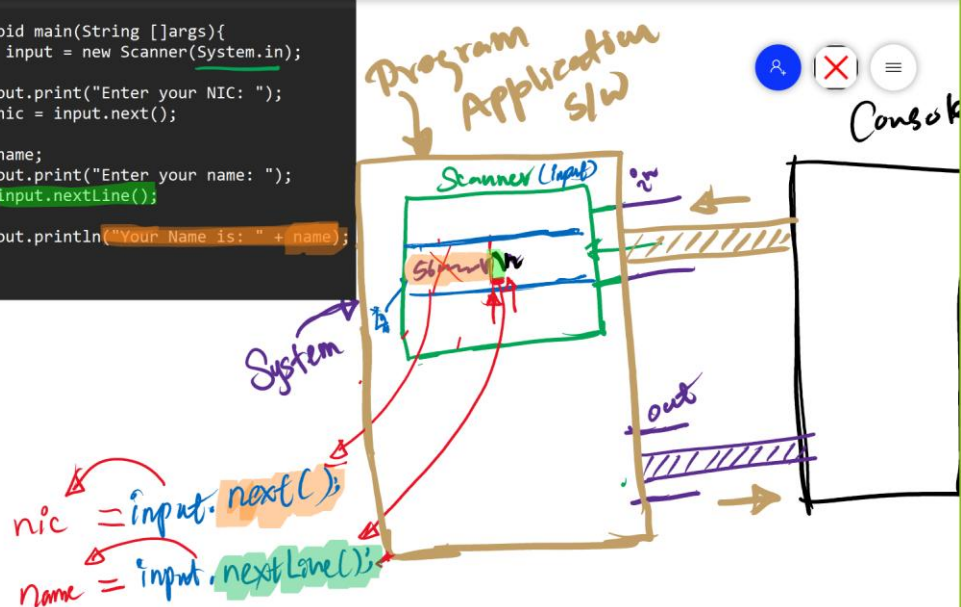


```
class Example{
    public static void main(String []args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your NIC: ");
        String nic = input.next();

        String name;
        System.out.print("Enter your name: ");
        name = input.nextLine();

        System.out.println("Your Name is: " + name);
    }
}
```



////////////////////////////////////

////////////////////////////////////

}

[illegible]



(1) a-z A-Z 0-9 _- only

Rules

(14) Case Sensitive

int a; ✓
int A; ✓

method n

1st letter $\rightarrow$ upper case

Converting

(8) Camel Case / Snake Case 

Q93

```
class Example{
```

```
int x;
```

```
int name;
```

```
int x1;
```

```
//int 1x;
```

```
int ab_;
```

```
int ____;
```

```
int x;
```

```
// int ; // Java 9 - 21 versions Error, 8- OK |
```

22+ Ok.

```
int ab$;
```

```
//int x#;
```

```
int $ab;
```

```
int name;
```


Q95 System.out.printf();

=====

```
import java.util.*;
class Example{
    public static void main(String []args){
        System.out.printf("Hello World!\n");

        int x = 100;
        System.out.println(x); // 100

        //System.out.printf(x); // Compile Error

        System.out.printf("%d", x); // 100
    }
}
```

////////////////////////////////////

Q96

```
import java.util.*;
class Example{
    public static void main(String []args){
        System.out.printf("Hello World!\n");

        int x = 100;
        System.out.println(x); // 100

        //System.out.printf(x); // Compile Error

        String name = "Dinuka";

        System.out.printf("%d%s", x, name); //
100Dinuka
    }
}
```

```
import java.util.*;
```

```
// Integer -> %d
// Floating Point -> %f
// String -> %s
//
```

//////////////////////////////////// Data Representation in Computer Memory ////////////////////////////////////

=====



D RAM - Random Access Memory

1MB

main memory

working memory

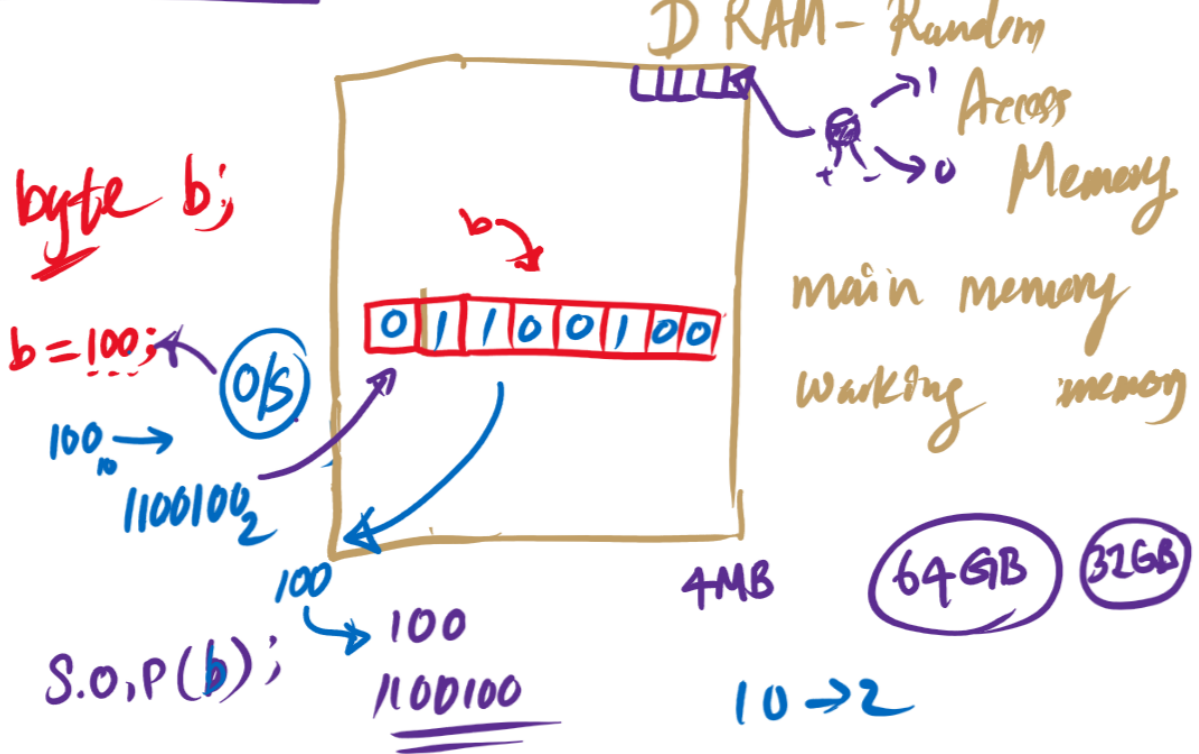
4MB

64GB

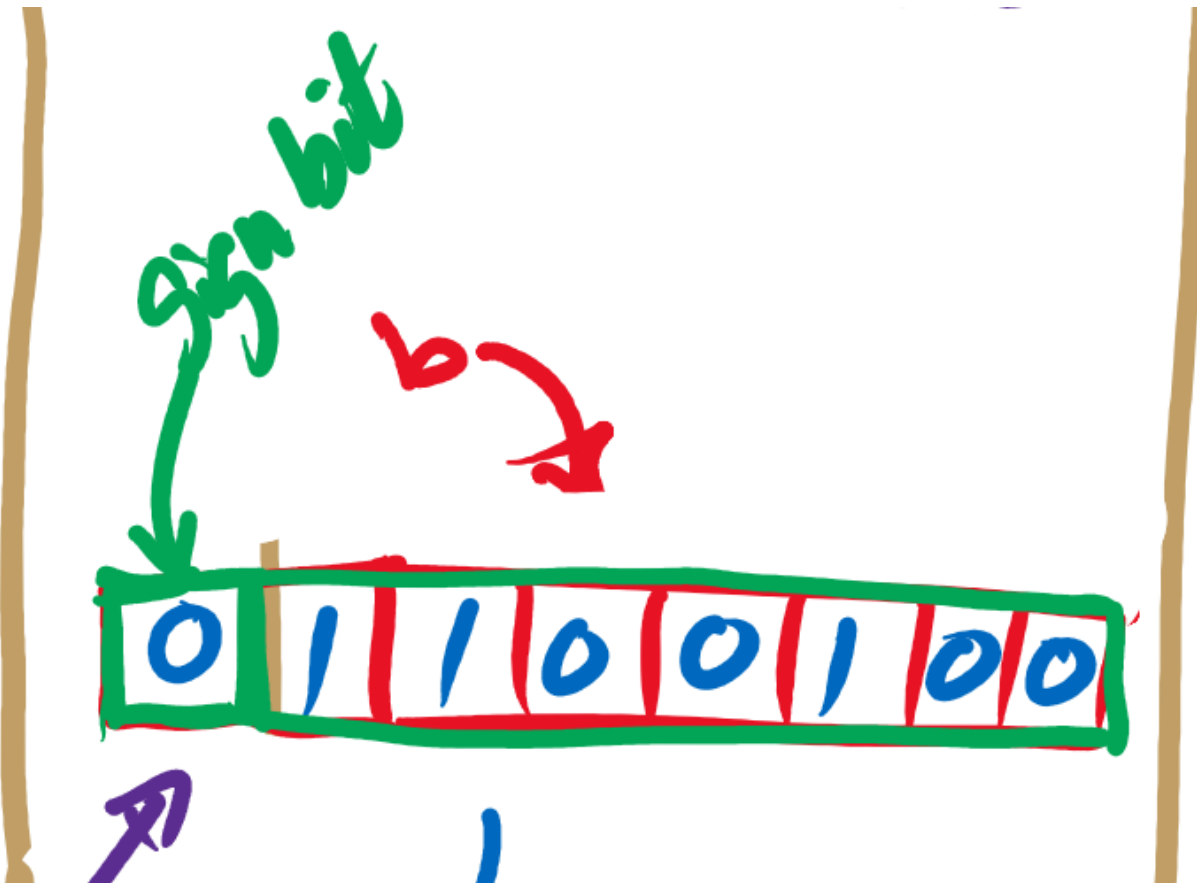
32GB

////////////////////////////////////

}

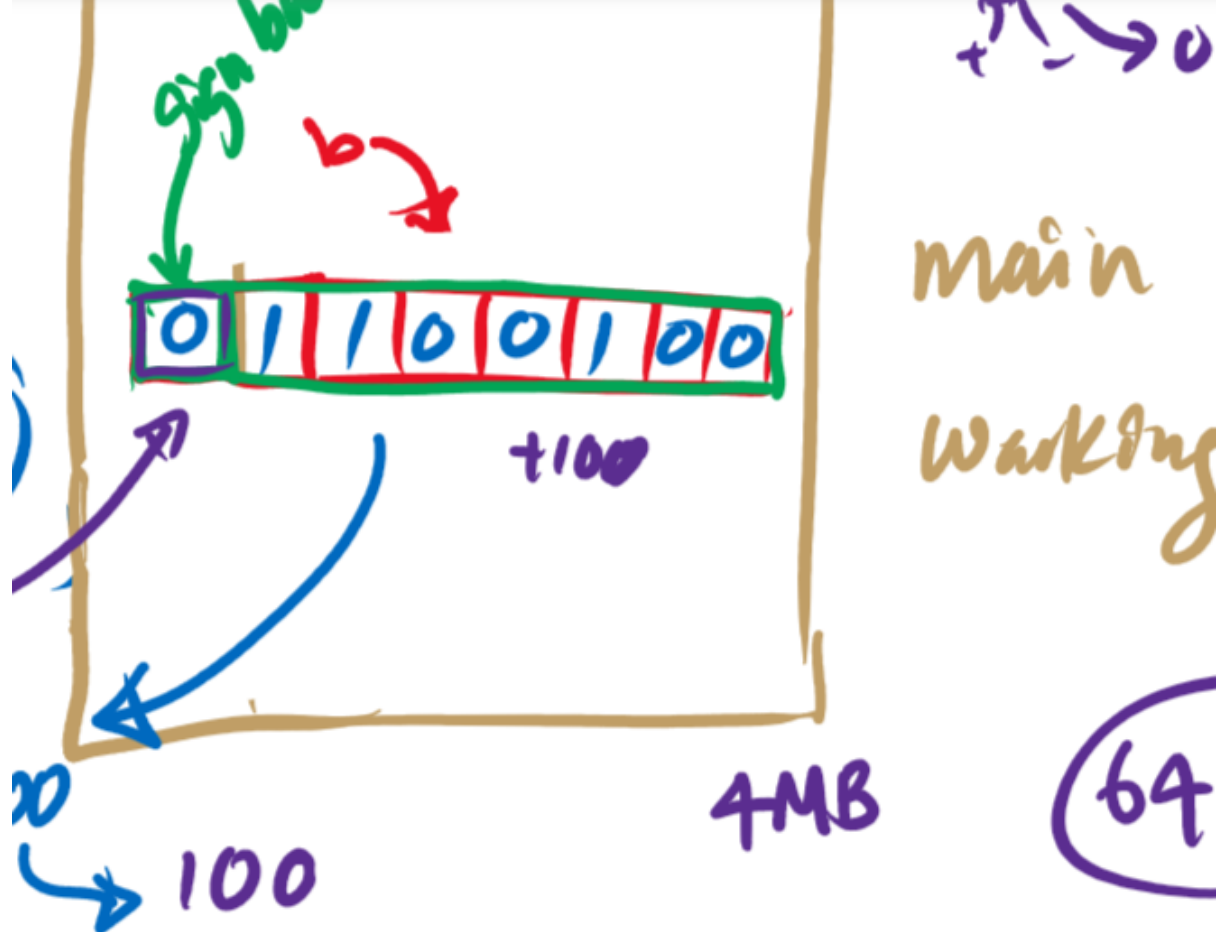


Q102



Q103 Negative Values

=====



1100100

10 → 2

111100100

⇒ ?

0 → 00000000

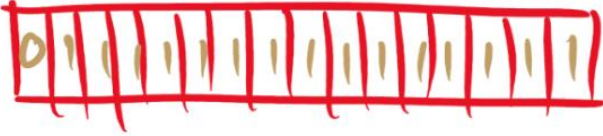
+0?

00000000 ⇒ ? -0?

[illegible]

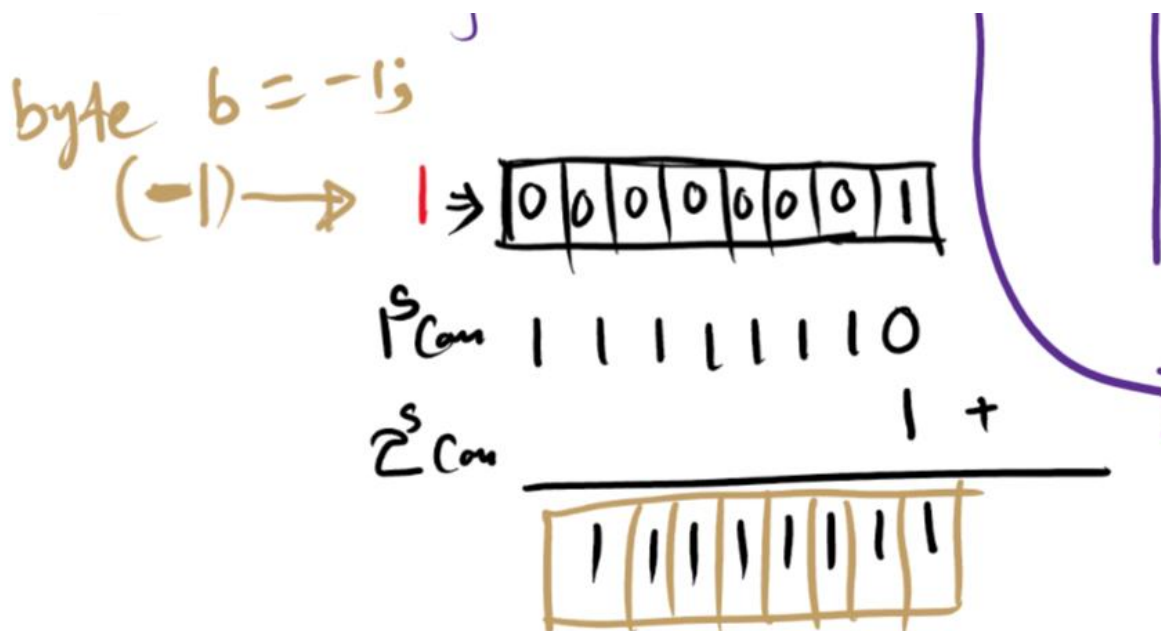
=====



byte	8	$2^7 - 1$	
Short	16	$2^{15} - 1$	
int	32	$2^{31} - 1$	
long	64	$2^{63} - 1$	
float	32		
double	64		
char	16		
boolean	1		

////////////////////////////////////

Q106

Short $s = -1$ 

Q107 byte Min value

byte $b = -128;$

$-128 \rightarrow 128$ 10000000

1 comp 01111111 2^8 con
 2^8 comp 1 +

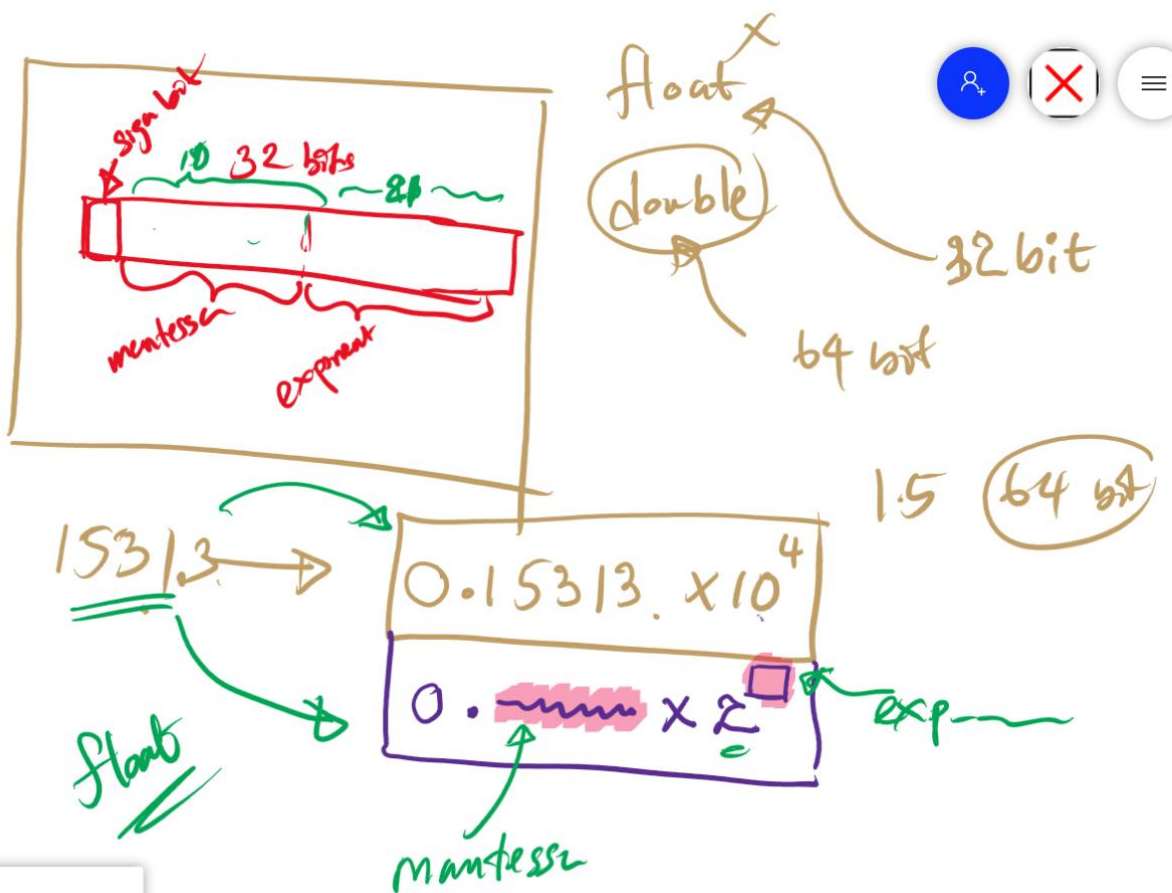


Sign Bit

Blue violet pen

		max	min
byte	8	$2^7 - 1$	-2^7
Short	16	$2^{15} - 1$	-2^{15}
int	32	$2^{31} - 1$	-2^{31}
long	64	$2^{63} - 1$	-2^{63}

Q108 Floating Point Numbers



Blue violet pen



Q109

```
import java.util.*;
class Example{
    public static void main(String []args){
        byte bMin = Byte.MIN_VALUE;
        byte bMax = Byte.MAX_VALUE;

        short sMin = Short.MIN_VALUE;
        short sMax = Short.MAX_VALUE;

        int xMin = Integer.MIN_VALUE;
        int xMax = Integer.MAX_VALUE;

        long lMin = Long.MIN_VALUE;
        long lMax = Long.MAX_VALUE;
    }
}
```

//

Q110

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String []args){
```

```
        // long - 64bits, float - 32bits
```

```
        float f = Long.MAX_VALUE; // ??
```

```
        System.out.println("long max value: " +  
Long.MAX_VALUE);
```

```
        System.out.println("f: " + f);
```

```
        long l = f; // Error
```

```
    }
```

```
}
```

//

Q112 Character Representation

=====

```

import java.util.*;
class Example{
    public static void main(String []args){
        char ch = 65;

        System.out.println(ch);
    }
}

```

Character 16 bits

'A' ASCII →

 → Integer
 value
 65



////////////////////////////////////

Q113

```
import java.util.*;
```

```
class Example{
    public static void main(String []args){
        char ch = 'h';

        System.out.println(ch);
        System.out.println(ch + 0); // ASCii Value of ch
: 'h'

        System.out.println('A' + 0);

        int x = ch;
        System.out.println(x);
    }
}
```

////////////////////////////////////

Q114 Boolean Representation

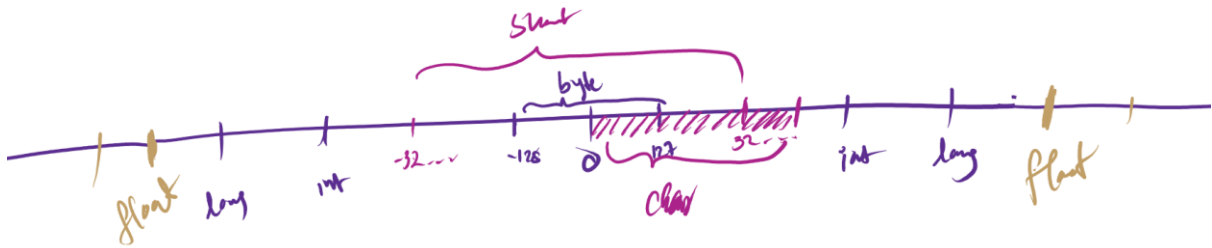
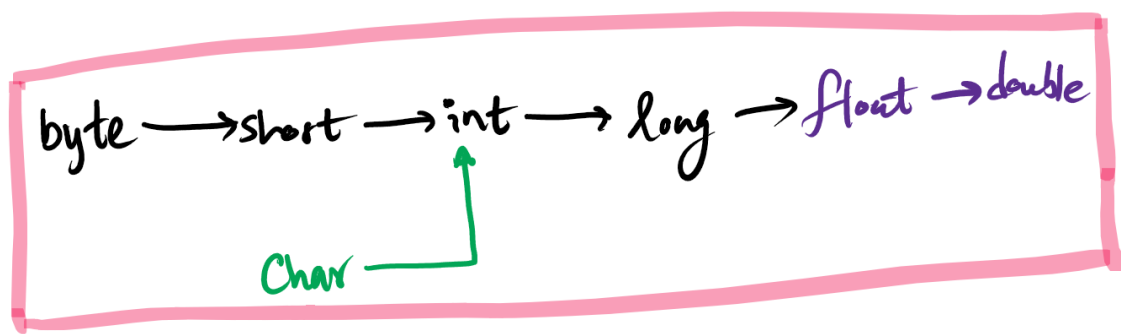
=====

Boolean 1 bit
□
1 → true
0 → false

////////////////////////////////////
Conversion and Casting
////////////////////////////////////

Q115

```
class Example{  
    public static void main(String []args){  
        byte b = 100;  
        short s = b;  
        int x = s;  
        long l = x;  
  
        float f = l;  
        double d = f;  
  
        char ch = 'A';  
  
        byte b2 = ch; // error  
        short s2 = ch; // error  
        int y = ch; // legal  
    }  
}
```



////////////////////////////////////

Q116

```
import java.util.*;
```

```
class Example{
```

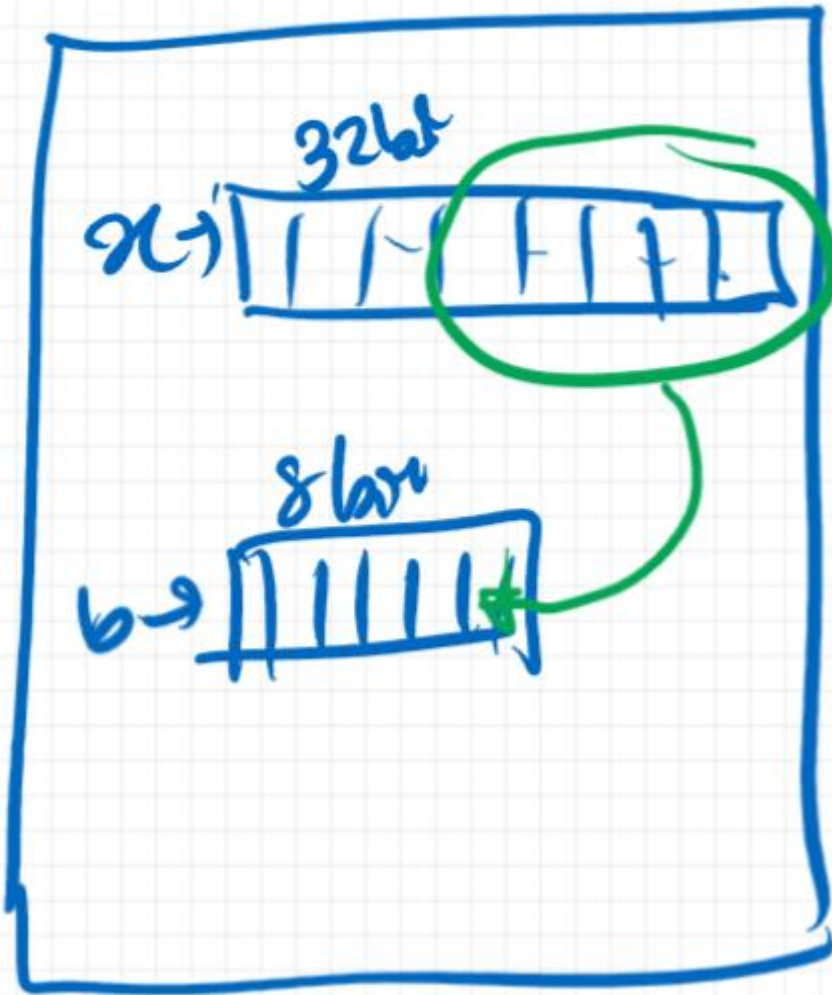
```
    public static void main(String []args){
```

```
        int x = 100;
```

```
        byte b = (byte) x;
```

```
    }
```

```
}
```



}

2026-03-15

//

Q118 Exercise

=====

```
class Example{
    public static void main(String args[]){
        int x = 12;
        int y = 15;
        System.out.println(x + " : " + y); // 12 : 15

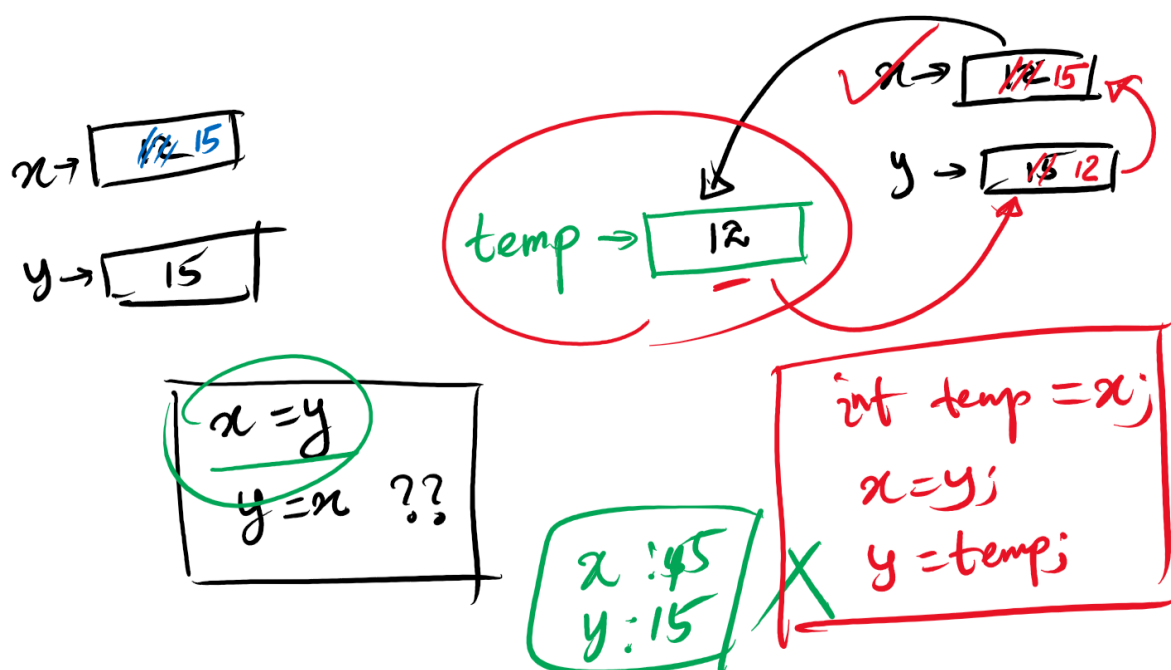
        //
        //
        //

        System.out.println(x + " : " + y); // 15 : 12
    }
}
```

//

Q119 From 118

=====



```

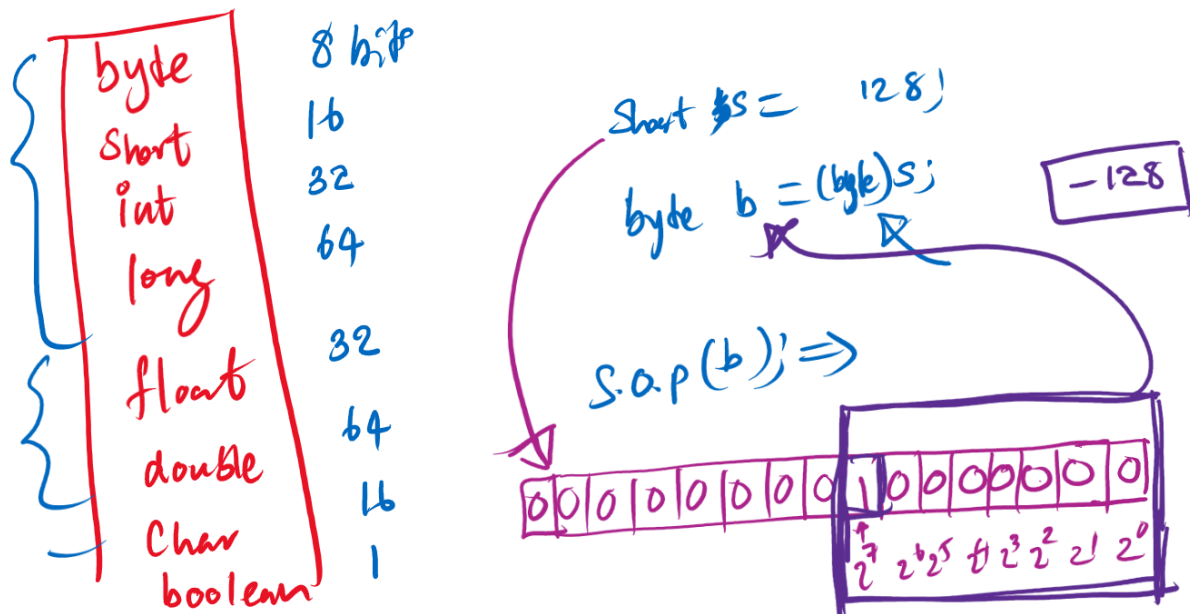
class Example{
    public static void main(String args[]){
        int x = 12;
        int y = 15;
        System.out.println(x + " : " + y); // 12 : 15

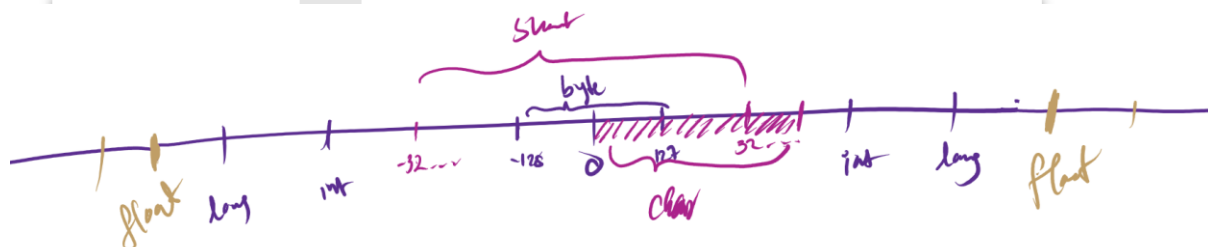
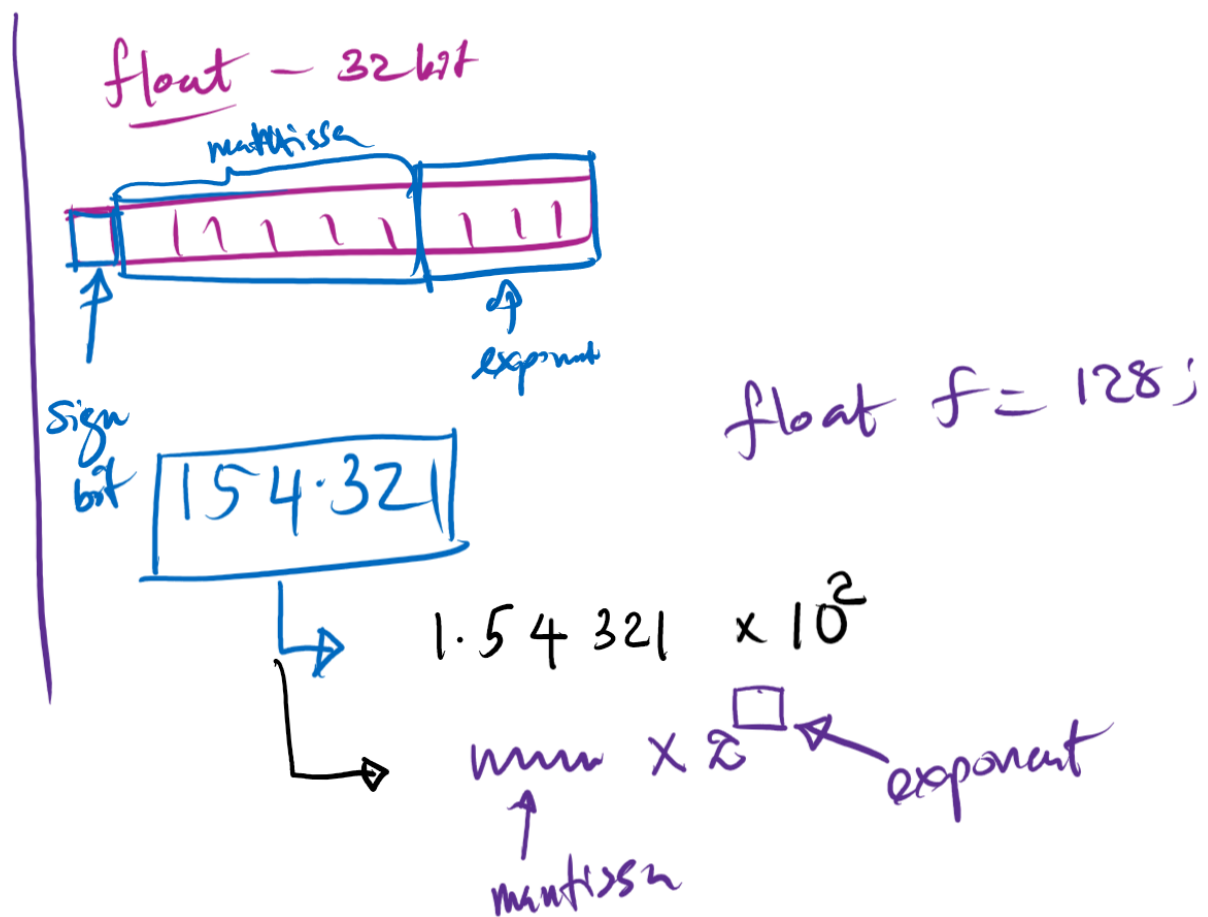
        int temp = x;
        x = y;
        y = temp;
        System.out.println(x + " : " + y); // 15 : 12
    }
}

```

////////////////////////////////////

Q120



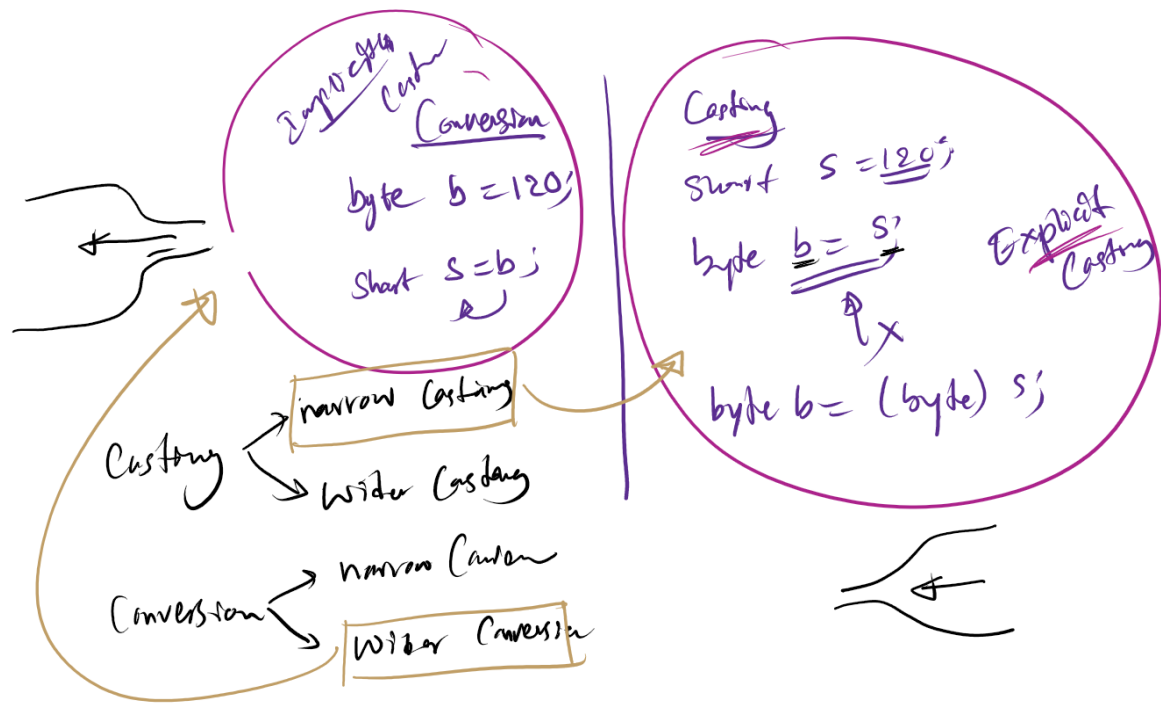


////////////////////////////////////

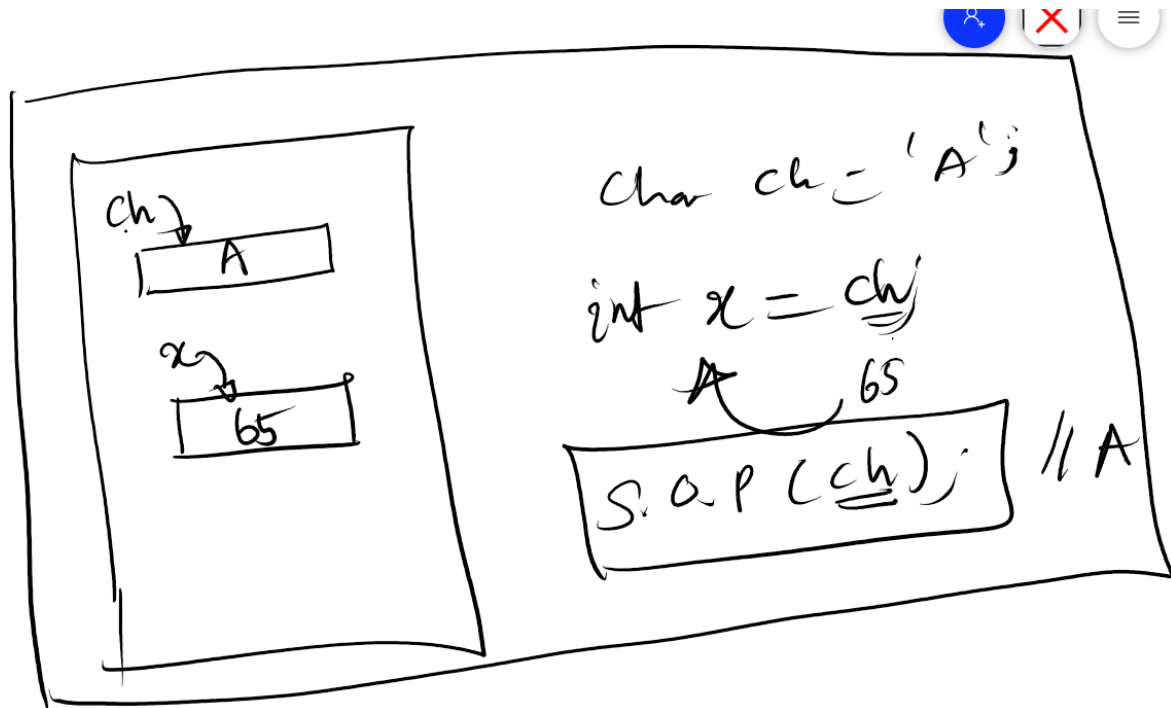
Q121

```
class Example{
    public static void main(String args[]){
        float f = 128; //
        int x = (int) f; //
        System.out.println(x); // 128
    }
}
```

Q122



Q123



```
class Example{
    public static void main(String args[]){
        char ch = 'A';
        int x = ch;

        System.out.println(ch); // A
    }
}
```

```
    }  
}
```

//

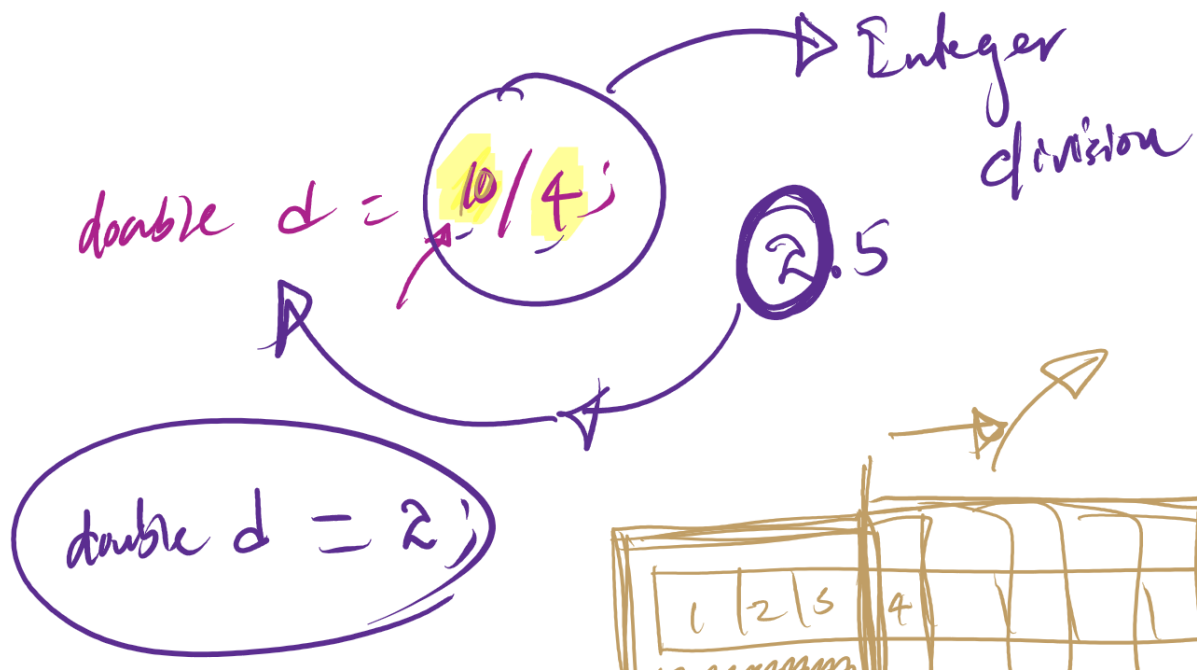
Q124

```
class Example{  
    public static void main(String args[]){  
        char ch = 'A';  
        int x = ch; // Wider Conversion  
  
        System.out.println(ch); // A  
        System.out.println(x); // 65  
    }  
}
```

//

Q125

```
class Example{  
    public static void main(String args[]){  
        double d = 10 / 4;  
        System.out.println(d); // 2.0  
    }  
}
```



////////////////////////////////////

Q126

```
class Example{
    public static void main(String args[]){
        double d1 = 10.0 / 4;
        double d2 = 10 / 4.0;

        double d3 = 10.0 / 4.0;

        System.out.println(d1); // 2.5
        System.out.println(d2); // 2.5
        System.out.println(d3); // 2.5
    }
}
```

////////////////////////////////////

Q127

```
class Example{
    public static void main(String args[]){
        double d = (double)(10 / 4);

        System.out.println(d); // 2.0
    }
}
```

//

Q128

```
class Example{
    public static void main(String args[]){
        double d1 = (double) 10 / 4;
        double d2 = 10 / (double) 4; // Wider Casting

        System.out.println(d1); // 2.5
        System.out.println(d2); // 2.5
    }
}
```

//

Q129

```
class Example{
    public static void main(String args[]){
        int total = 100;
        int n = 24;

        double avg = (double) total / n; // Wider Casting
        System.out.println(avg); // 4.0
    }
}
```

Arithmetic Operations



Arithmetic Operations

unary

+ Plus Sign
- minus Sign

++

--

binary

+ addition

- Subtraction

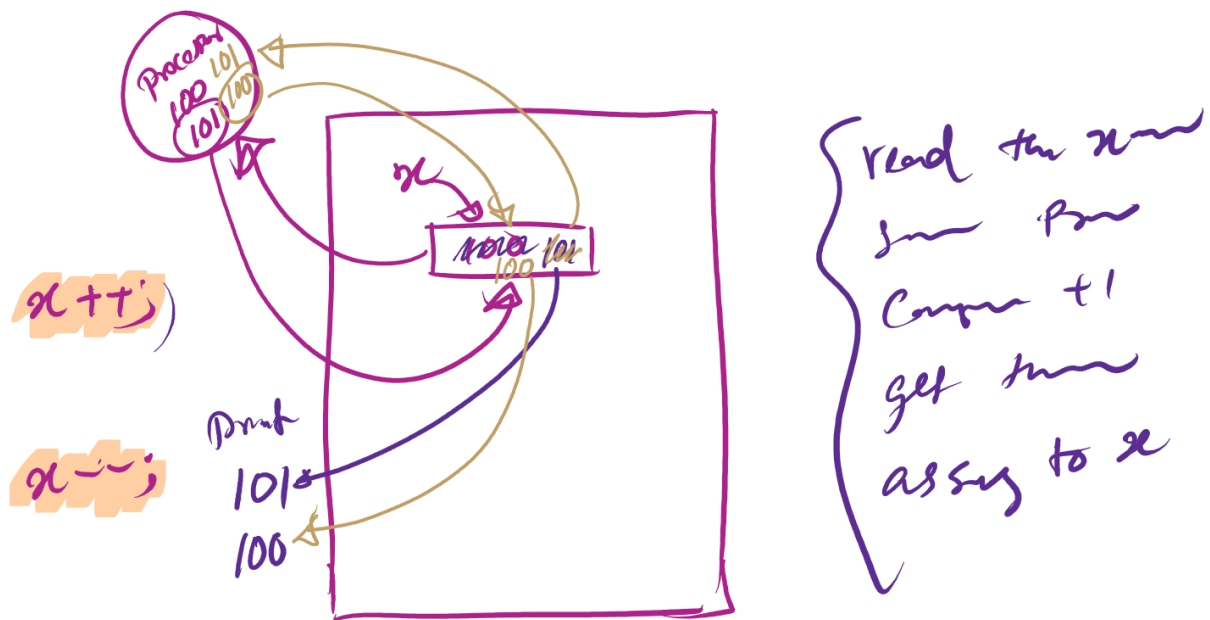
/ division

* multiplication

% modular / remainder

Q130 Unary Operations

```
class Example{  
    public static void main(String args[]){  
        int x = 100;  
        System.out.println(+x);    //100  
        System.out.println(+100); //100  
  
        System.out.println(-x); //-100  
        System.out.println(-100); //-100  
    }  
}
```

//

Q133

```
class Example{
    public static void main(String args[]){
        int x = 100;
        System.out.println(x++); //
        System.out.println(x--); //
    }
}
```

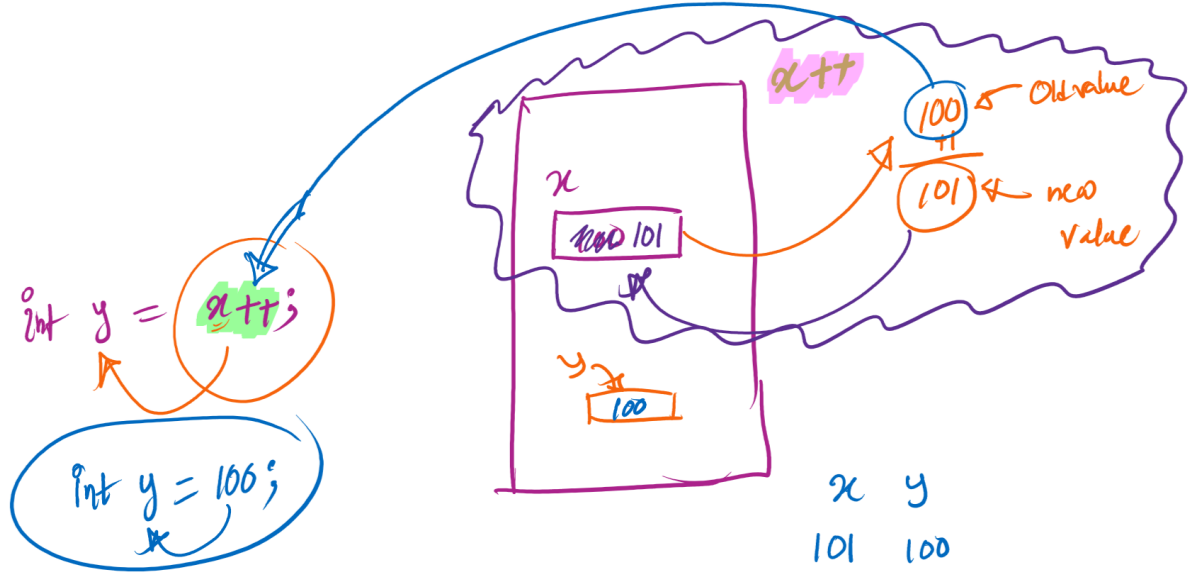
//

Q134

```
class Example{
    public static void main(String args[]){
        int x = 100;

        int y = x++;

        System.out.println(x + " " + y); // 101 100
    }
}
```



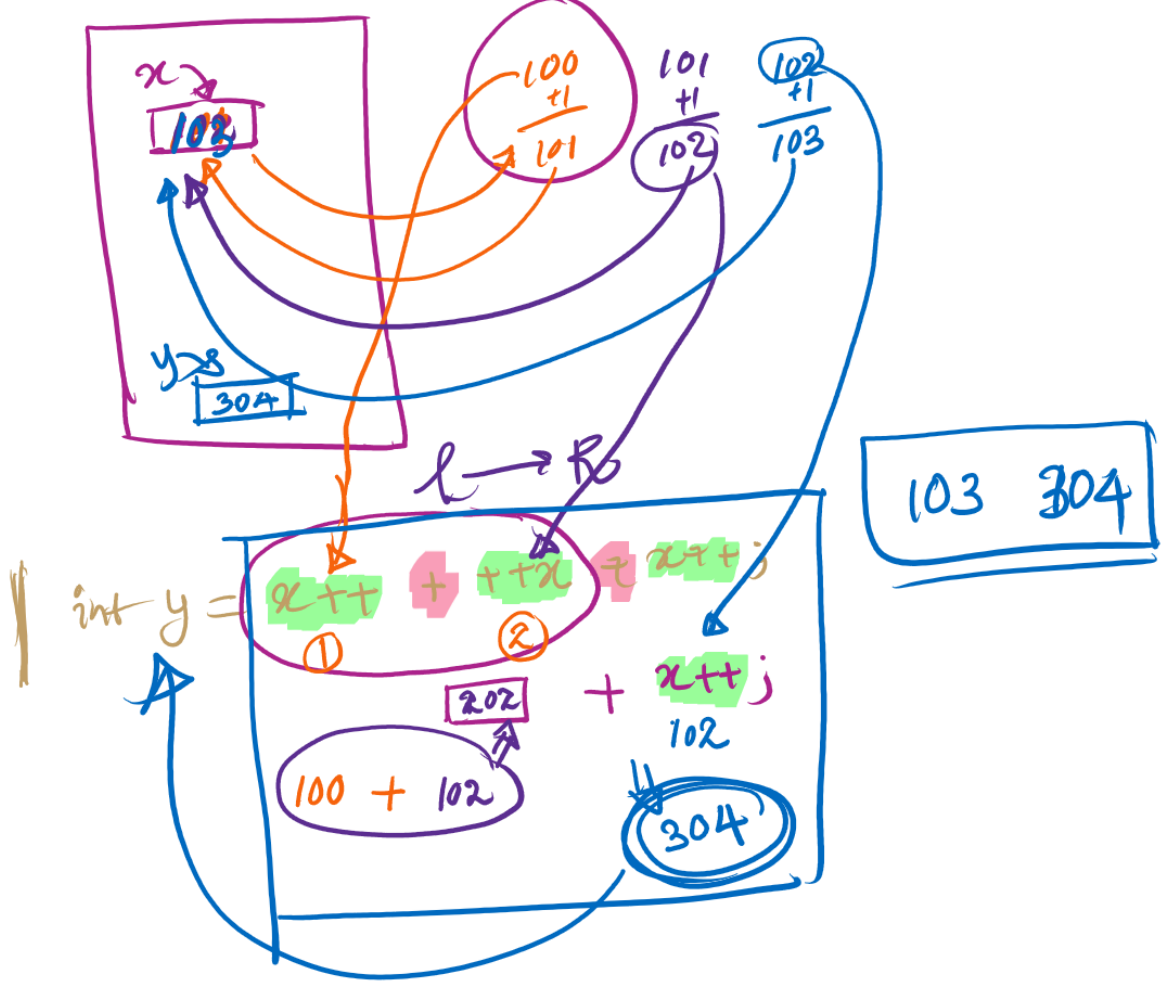
////////////////////////////////////

Q135

```
class Example{
    public static void main(String args[]){
        int x = 100;

        int y = x++ + ++x + x++;

        System.out.println(x + " " + y); //
    }
}
```



Operator Precedence

()

$l \rightarrow R$

++ --

* / %

+ , -

Q137

Input length: = 18857

mm = 1885751 =

1 Km 885m 75cm 1mm

1000530 mm
1 Km 0 m 53cm 0 mm

Operator

()

++ --

* / %

+ , -

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input length: ");
```

```
        int length = input.nextInt();
```

```
        int km, m, cm, mm;
```

```
        km = length / 1_000_000;
```

```
        length = length % 1_000_000;
```

```
        m = length / 1000;
```

```
        length = length % 1000;
```

Activate

```
cm = length / 10;
```

```
mm = length % 10;
```

```
System.out.println(km +"km, " + m + "m, " + cm  
+ "cm, " + mm + "mm ");
```

```
}
```

```
}
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q138

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input amount: ");
```

```
        int amount = input.nextInt();
```

```
        int r5000, r1000, r500, r100, r50, r20, r10, r5,  
r2, r1;
```

```
}
```

```
}
```

```
=====
```

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input amount: ");
```

```
        int amount = input.nextInt();
```

```
int r5000, r1000, r500, r100, r50, r20, r10, r5,  
r2, r1;
```

```
    r5000 = amount / 5000;  
    amount = amount % 5000;  
  
    }  
}
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q139 Operator Assignment

```
=====
```

```
import java.util.*;  
class Example{  
    public static void main(String args[]){  
        int x = 100;  
  
        x = x + 20;  
  
        x += 20; // Operator Assignment  
  
        x -= 10;  
  
        x /= 2;  
  
        x *= 3;  
  
        x %= 2;  
  
        System.out.println("x : " + x);  
    }  
}
```


////////////////////////////////////
Q140

import java.util.*;

class Example{

public static void main(String args[]){

int x = 100;

x = x + 20;

x += 20; // Operator Assignment

x -= 10;

x /= 2;

x *= 3;

x %= 2;

System.out.println("x : " + x);

}

}

Q141

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        short s = 100;
```

```
        byte b = 5;
```

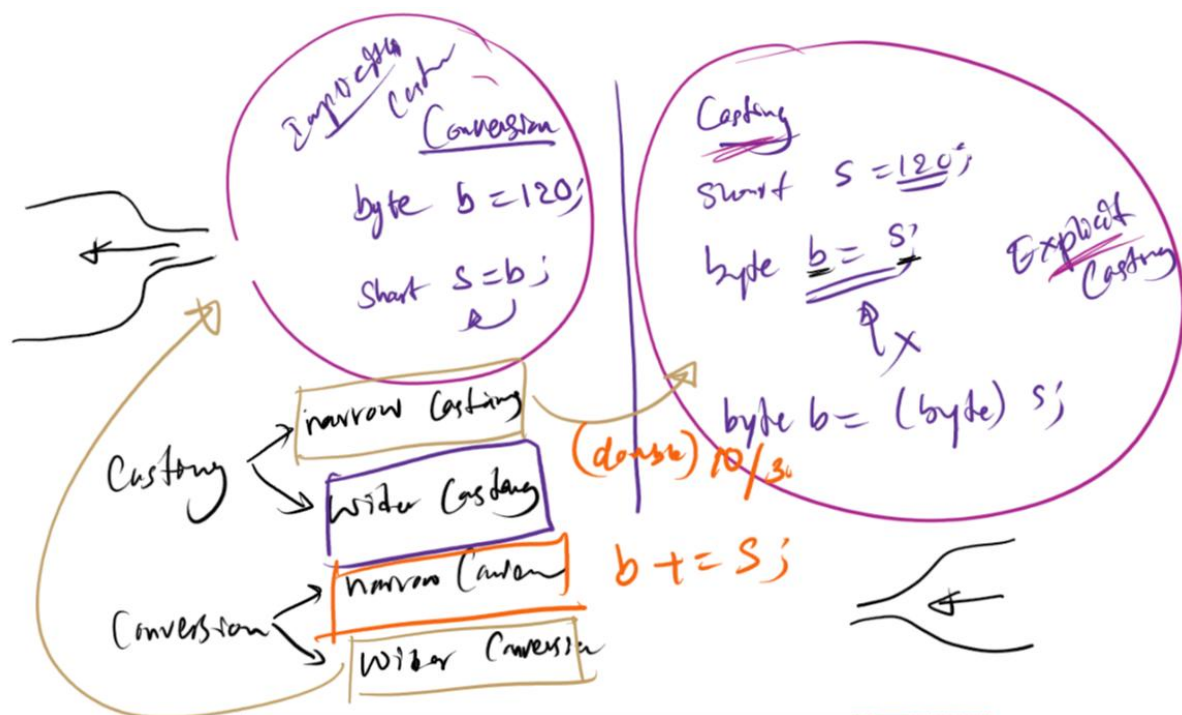
```
        // b = b + s; // Error
```

```
        b += s; // Narrow conversion
```

```
        System.out.println("b : " + b);
```

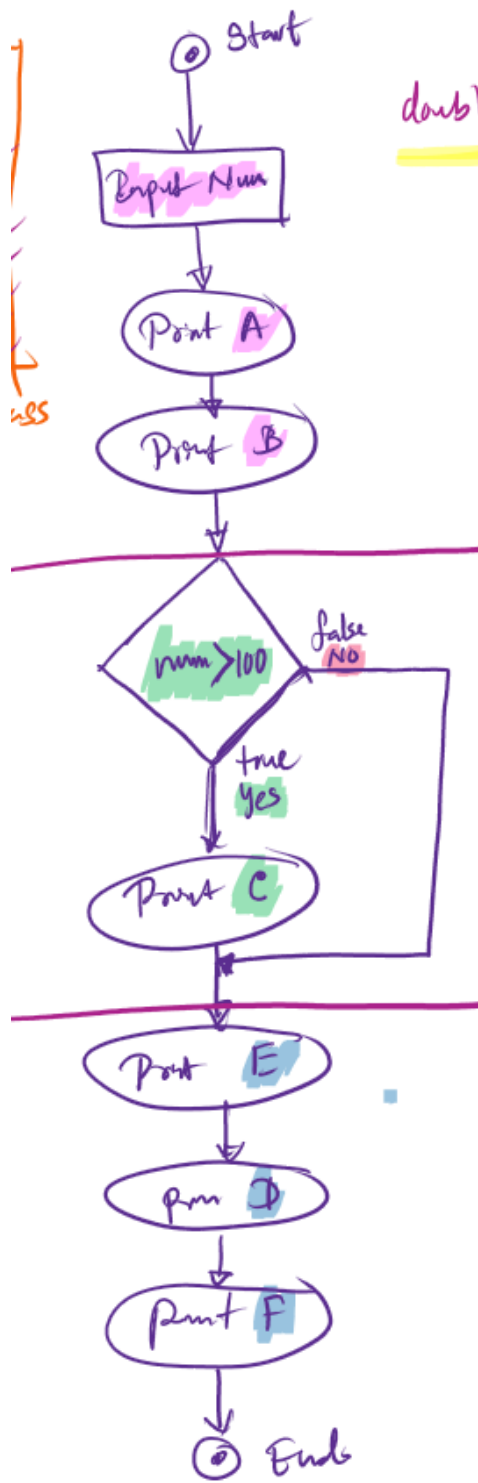
```
    }
```

```
}
```



////////////////////////////////////





////////////////////////////////////
Q142

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Number: ");
        int num = input.nextInt();

        System.out.println("A");
        System.out.println("B");
        System.out.println("C");
        System.out.println("D");
        System.out.println("E");
        System.out.println("F");
    }
}
```

////////////////////////////////////
Q143

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Number: ");
        int num = input.nextInt();

        System.out.println("A");
        System.out.println("B");
        if(num > 100) System.out.println("C");
        System.out.println("D");
        System.out.println("E");
        System.out.println("F");
    }
}
```

////////////////////////////////////

```
import java.util.*;
```

```
class Example{
```

```
public static void main(String args[]){
```

```
Scanner input = new Scanner(System.in);
```

```
System.out.print("Input Number: ");
```

```
int num = input.nextInt();
```

```
System.out.println("A");
```

```
System.out.println("B");
```

```
if(num > 100)
```

```
System.out.println("C");
```

```
System.out.println("D");
```

```
System.out.println("E");
```

```
System.out.println("F");
```

}

}

////////////////////////////////////

Q145

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input Number: ");
```

```
        int num = input.nextInt(); // 99
```

```
        System.out.println("A");
```

```
        System.out.println("B");
```

```
        if(num > 100)
```

```
            System.out.println("C");
```

```
            System.out.println("D");
```

```
        System.out.println("E");
```

```
        System.out.println("F");
```

```
    }
```

```
}
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q146

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input Number: ");
```

```
        int num = input.nextInt(); // 99
```

```
        System.out.println("A");
```

```
        System.out.println("B");
```

```
        if(num > 100)
```

```
            {System.out.println("C");
```

```
            System.out.println("D");}
```

```
        System.out.println("E");
```

```
        System.out.println("F");
```

```
}
```

////////////////////////////////////

```
import java.util.*;
```

```
public static void main(String args[]){
```

```
System.out.print("Input Number: ");
```

```
int num = input.nextInt(); // 99
```

```
System.out.println("A");
```

```
System.out.println("B");
```

```
if(num > 100){
```

```
System.out.println("C");
```

```
System.out.println("D");
```

}

```
System.out.println("E");
```

```
System.out.println("F");
```

}

////////////////////////////////////

Q148

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input Number: ");
```

```
        int num = input.nextInt(); //
```

```
        System.out.println("A");
```

```
        System.out.println("B");
```

```
        if(num > 100) {
```

```
            System.out.println("C");
```

```
        }
```

```
        if(num < 100) {
```

```
            System.out.println("D");
```

```
            System.out.println("E");
```

```
        }
```

```
        System.out.println("F");
```

```
    }
```

```
}
```

////////////////////////////////////

////////////////////////////////////

Q151

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Salary: ");
        int salary = input.nextInt();

        if(salary < 100_000) { // if block
            salary += 5000;
        }

        System.out.println("New Salary: " + salary);
    }
}
```

////////////////////////////////////

Q152

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Average Marks: ");
        double avg = input.nextDouble();

        if(avg > 50) System.out.println("Pass");

        if(avg < 50) System.out.println("Fail");
    }
}
```

Q153

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Average Marks: ");
        double avg = input.nextDouble();

        if(avg > 50) {
            System.out.println("Pass");
        }

        if(avg < 50) {
            System.out.println("Fail");
        }
    }
}
```

Java Flow Control "if/else"

////////////////////////////////////

Q154

```
import java.util.*;
```

```
class Example{
```

```
public static void main(String args[]){
```

```
Scanner input = new Scanner(System.in);
```

```
System.out.print("Input Average Marks: ");
```

```
double avg = input.nextDouble();
```

```
if(avg > 50) {
```

```
System.out.println("Pass");
```

```
} else {
```

```
System.out.println("Fail");
```

}

}

}

////////////////////////////////////

Q155 Exercise I

```
import java.util.*;
```

```
class Example{
```

```
public static void main(String args[]){
```

```
Scanner input = new Scanner(System.in);
```

```
System.out.print("Input Salary: ");
```

```
double salary = input.nextDouble();
```

```
//less than 100000 -> 10% increment
```

```
//greater than 100000 -> 5% increment
```

```
System.out.println("New Salary: " + salary);
```

}

}

////////////////////////////////////

Q156 From Q155

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Salary: ");
        double salary = input.nextDouble();

        if(salary < 100_000) {
            // 10% increment
        } else {
            // 5% increment
        }

        if(salary > 100_000) {
            // 5% increment
        } else {
            // 10% increment
        }

        System.out.println("New Salary: " + salary);
    }
}
```

////////////////////////////////////

Q157 From Q156

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Salary: ");
        double salary = input.nextDouble();

        if(salary > 100_000) {
            // 5% increment
            salary = salary + (salary * 5/100);
        } else {
            // 10% increment
            salary = salary + (salary * 10/100);
        }

        System.out.println("New Salary: " + salary);
    }
}
```

////////////////////////////////////

Q158 From Q157

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Salary: ");
        double salary = input.nextDouble();

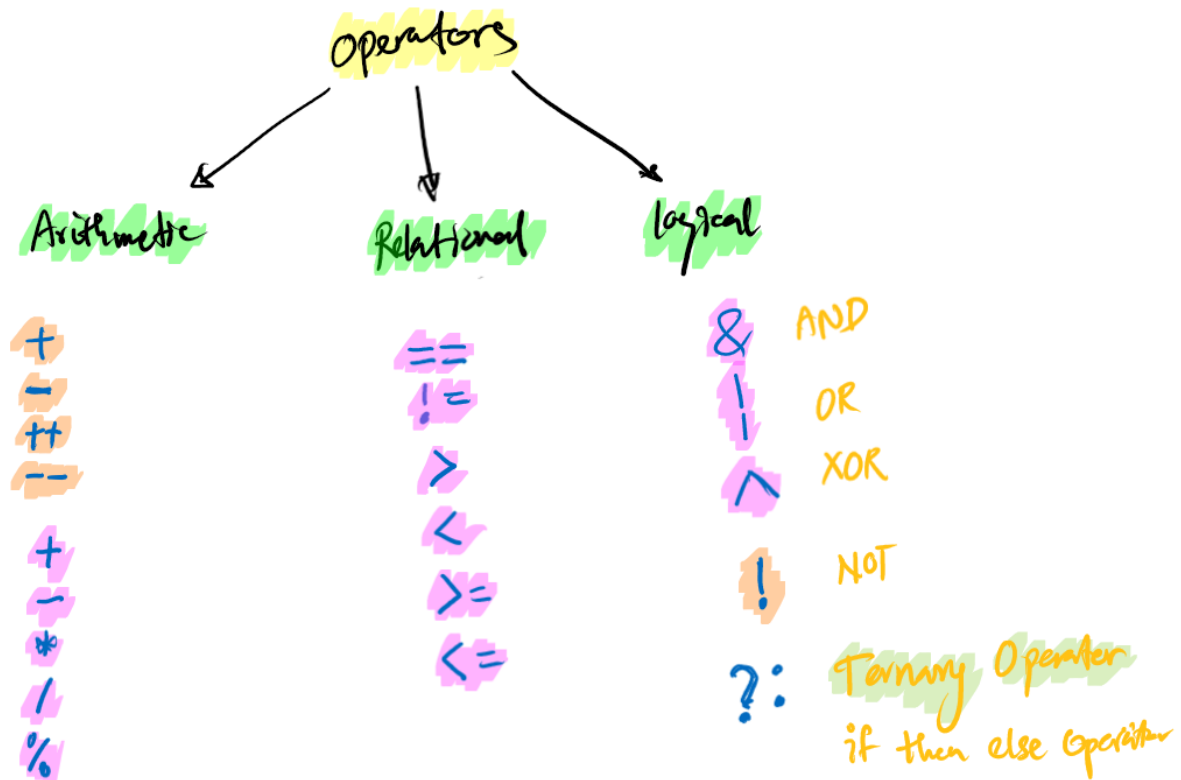
        if(salary > 100_000) {
            // 5% increment
            // salary = salary + (salary * 5/100);
            // salary = salary * 105/100;
            salary *= 1.05;
        } else {
            // 10% increment
            // salary = salary + (salary * 10/100);
            salary *= 1.1;
        }

        System.out.println("New Salary: " + salary);
    }
}
```

////////////////////////////////////

Relational Operators

Q159



XOR

Input		Output
A	B	X
true	true	false
true	false	true
false	true	true
false	false	false

Q160

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        boolean b1 = true;
```

```
        boolean b2 = false;
```

```
        System.out.println(100 == 100); // true
```

```
        System.out.println(100 != 100); // false
```

```
        System.out.println(100 != 101); // true
```

```
        System.out.println(!b1); // false
```

```
        System.out.println(!b2); // true
```

```
        // System.out.println(!100); // Illegal
```

```
        System.out.println(b1 == b2); // false
```

```
        System.out.println(100 >= 101); // false
```

```
        // System.out.println(b1 > b2); // Illegal
```

```
        boolean b3 = 100 >= 101;
```

```
        System.out.println(!(100>=101)); // true
```

```
    }
```

```
}
```

////////////////////////////////////

Q161

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input a: ");
```

```
        int a = input.nextInt();
```

```
        System.out.print("Input b: ");
```

```
        int b = input.nextInt();
```

```
        System.out.print("Input c: ");
```

```
        int c = input.nextInt();
```

```
        System.out.print("Input d: ");
```

```
        int d = input.nextInt();
```

```
        System.out.print("Input e: ");
```

```
        int e = input.nextInt();
```

```
        int max;
```

```
        System.out.println("Maximum Value: " + max);
```

```
    }
```

```
}
```

////////////////////////////////////

Q162 From Q163

=====

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input a: ");
```

```
        int a = input.nextInt();
```

```
        System.out.print("Input b: ");
```

```
        int b = input.nextInt();
```

```
        System.out.print("Input c: ");
```

```
        int c = input.nextInt();
```

```
        System.out.print("Input d: ");
```

```
        int d = input.nextInt();
```

```
        System.out.print("Input e: ");
```

```
        int e = input.nextInt();
```

```
        int max = a;
```

```
        if(max < b) {
```

```
            max = b;
```

```
        }
```

```
        // checked a, b
```

```
        if(max < c) {
```

```
            max = c;
```

```
        }
```

```
        if(max < d) {
```

```
            max = d;
```

```
        }
```

```
        if(max < e) {
```

```
            max = e;
```

}

System.out.println("Maximum Value: " + max);

}

}

//

Q163

import java.util.*;

class Example{

public static void main(String args[]){

int x=99, y = 100, z=101;

System.out.println(x == y); // 99 == 100 false

System.out.println(x++ == y); // 99 == 100 false

System.out.println(++x == y); // 101 == 100

false

System.out.println(x == y--); // 101 == 100 false

System.out.println(x == --y); // 101 == 98 false

System.out.println(x++ == --y); // 101 == 97

false

}

}

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x=99, y = 100, z=101;

        System.out.println(x == y); // 99 == 100 false
        System.out.println(x++ == y); // 99 == 100 false
        System.out.println(++x == y); // 101 == 100 false
        System.out.println(x == y--); // 101 == 100 false
        System.out.println(x == --y); // 101 == 98 false
        System.out.println(x++ == --y); // 101 == 97 false
    }
}
```

x y
99 100
100
101

$x == y$

$99 == 100 \rightarrow \text{false} \checkmark$

$x++ == y$

$99 == 100 \rightarrow \text{false}$

$++x == y$

$101 == 100 \rightarrow \text{false}$

//

Q164 "==" vs "=="

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x=99, y = 100, z=101;

        System.out.println(x=y); // 100
        System.out.println(x=100); // 100

        // System.out.println(x = y == z); // Illegal
        System.out.println((x=y) == z); // false

        // System.out.println((x=y) == z = 100); // Illegal
        System.out.println((x=y) == (z = 100)); // true
    }
}
```

//

Q165

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input marks: ");
        double avg = input.nextDouble();
```

```
        System.out.println("Grade A"); // 75 - 100
        System.out.println("Grade B"); // 65 - 75
        System.out.println("Grade C"); // 50 - 65
        System.out.println("Grade F"); // <50
```

```
    }
}
```

//

Q166 // Logical Issue

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input marks: ");
        double avg = input.nextDouble();

        if(avg >= 75) {
            System.out.println("Grade A"); // 75 - 100
        }
        if(avg >= 65){
            System.out.println("Grade B"); // 65 - 75
        }
        if(avg >=50){
            System.out.println("Grade C"); // 50 - 65
        } else {
            System.out.println("Grade F"); // <50
        }
    }
}
```

Q167 From Q166

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input marks: ");
        double avg = input.nextDouble();

        if(avg >= 75) {
            System.out.println("Grade A"); // 75 - 100

        } else
        if(avg >= 65){
            System.out.println("Grade B"); // 65 - 75
        } else
        if(avg >=50){
            System.out.println("Grade C"); // 50 - 65
        } else {
            System.out.println("Grade F"); // <50
        }
    }
}
```

if/else if

////////////////////////////////////

Q168 From Q169

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input marks: ");
        double avg = input.nextDouble();

        if(avg >= 75) {
            System.out.println("Grade A"); // 75 - 100
        } else if(avg >= 65){
            System.out.println("Grade B"); // 65 - 75
        } else if(avg >=50){
            System.out.println("Grade C"); // 50 - 65
        } else {
            System.out.println("Grade F"); // <50
        }
    }
}
```

////////////////////////////////////

Conditional Variable Initialization

```
////////////////////////////////////
```

Q169 Case I

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int y = 100;
        int x;

        if(y == 100) { // Illegal
            x = 200;
        }

        System.out.println(x);
    }
}
```

```
////////////////////////////////////
```

Q170 Case II

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int y = 100;
        int x;

        if(100 == 100) { // Legal
            x = 200;
        }

        System.out.println(x);
    }
}
```

```
////////////////////////////////////
```

Q171 Case III

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int y = 100;
        int x;

        if(y == 100) {
            x = 200;
        } else {
            x = 300;
        }

        System.out.println(x);
    }
}
```

Q172 Case IV

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int y = 100;
        int x;
        if(y == 100) {
            x = 200;
        } else {

        }
        System.out.println(x);
    }
}
```

Q173 Case V

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int y = 100;
        int x;
        if(y == 100) {
            x = 200;
        } else if(y > 150) {
            x = 300;
        }
        System.out.println(x);
    }
}
```

Q174 Case VI

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int y = 100;
        int x;
        if(y == 100) {
            x = 200;
        } else if(y > 150) {
            x = 300;
        } else {
            x = 400;
        }
        System.out.println(x);
    }
}
```

Logical Operators

logical

2

AND



OR

XOR



NOT

?

Ternary Operator

if then else operator

////////////////////////////////////

Q175

```
import java.util.*;
```

```
class Example{
```

```
public static void main(String args[]){
```

```
boolean b1 = true;
```

```
boolean b2 = false;
```

```
System.out.println(b1 & b2); // false
```

```
System.out.println(b1 & !b2); // true
```

```
System.out.println(b1 | b2); // true
```

```
System.out.println(b1 | !b2); // true
```

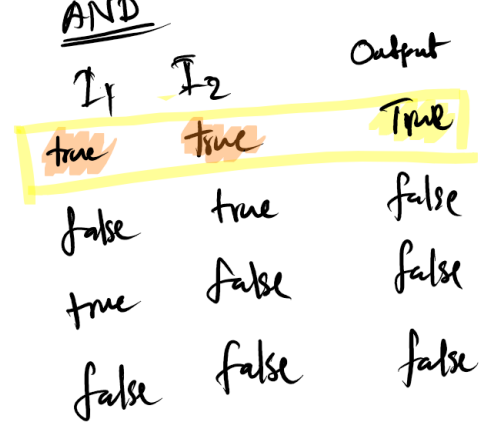
```
System.out.println(b1 ^ b2); // true
```

```
System.out.println(b1 ^ !b2); // false
```

}

}

"&" vs "&&"



Q176

```
class Example{
```

```
int x = 100, y = 101;
```

false

100

}

Q177 Short Circuit AND

```

=====
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 100, y = 101;

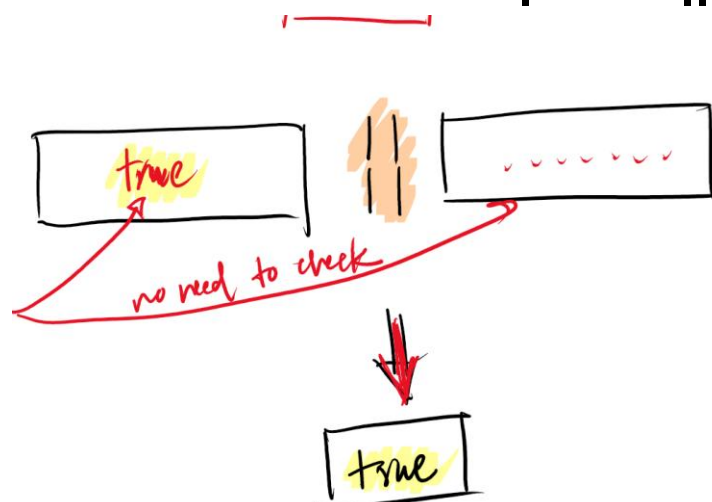
        //System.out.println(x++ & y--); // false
        System.out.println(x++ == 101 && y-- == 101);
        // false && ... => false

        System.out.println("x: " + x + " | y: " + y); // 101
101
    }
}

```

//

"|" vs "&&"



OR I ₁	I ₂	Output
false	false	false
true	false	true
false	true	true
true	true	true

//

```

import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 100, y = 101;

        //System.out.println(x++ & y--); // false
        System.out.println(x++ == 100 | y-- == 101); //
true

        System.out.println("x: " + x + " | y: " + y); // 101
100
    }
}

```

////////////////////////////////////

Q179 Short Circuit OR

=====

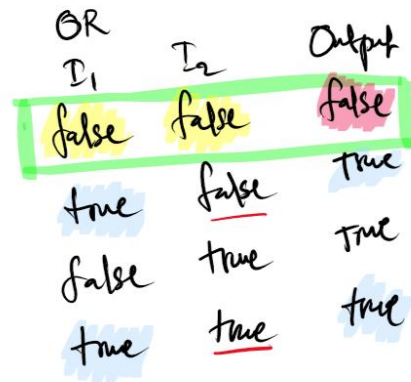
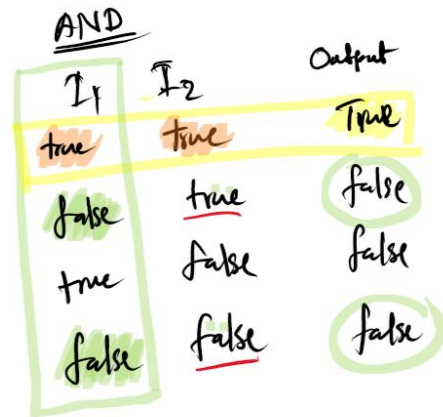
```

import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 100, y = 101;

        //System.out.println(x++ & y--); // false
        System.out.println(x++ == 100 || y-- == 101); //
true || ... => true

        System.out.println("x: " + x + " | y: " + y); // 101
101
    }
}

```

}

```
////////////////////////////////////
```

Q181 Exercise

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 100, y = 101;

        if (x == 101 || ((y = x) == 101)){
            x = y;
        }

        System.out.println("x: " + x + " | y: " + y); // x: |
y:
    }
}
```

```
////////////////////////////////////
```

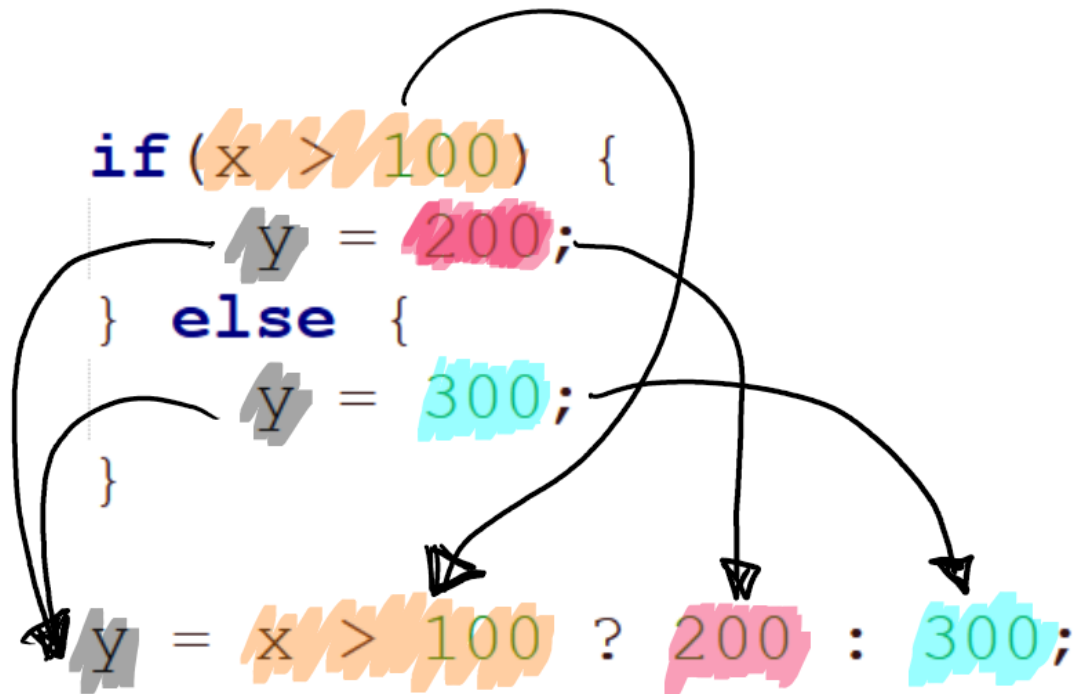
Turnery Operator ? :

```
////////////////////////////////////
```

Q182

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 99;
        int y;
        if(x > 100) {
            y = 200;
        } else {
            y = 300;
        }
        System.out.println("x: " + x + " | y: " + y); // x: |
y:
    }
}
```

Q183



```
import java.util.*;  
class Example{  
    public static void main(String args[]){  
        int x = 99;  
        int y;  
  
        if(x > 100) {  
            y = 200;  
        } else {  
            y = 300;  
        }  
  
        y = x > 100 ? 200 : 300;  
  
        System.out.println("x: " + x + " | y: " + y); // x: |  
        y:  
    }  
}
```

Q184

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 99;
        int y;

        y = x > 100 ? 200 : 300;
        System.out.println("x: " + x + " | y: " + y); // x: |
y:
    }
}
```

Q185

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 99;
        int y = x > 100 ? 200 : 300;
        System.out.println("x: " + x + " | y: " + y); // x: |
y:
    }
}
```

//

Q186

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 99;
        int y, z;
        if(x > 100) { // Cannot use Ternary Operator
            y = 100;
        } else {
            z = 200;
        }
        System.out.println("x: " + x + " | y: " + y); // x: |
y:
    }
}
```

//

Q187 Exercise

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Salary: ");
        double salary = input.nextDouble();
        if(salary > 100_000) {
            salary *= 1.05;
        } else {
            salary *= 1.1;
        }
        //
        System.out.println("New Salary: " + salary);
    }
}
```

Q188

```
class Example{
```

```
Scanner input = new Scanner(System.in);
```

```
double salary = input.nextDouble();
```

```
salary = salary * 1.05;
```

```
salary = salary * 1.1;
```

}

```
salary * 1.1;
```

```
System.out.println("New Salary: " + salary);
```

}

}

Q189

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input marks: ");
```

```
        double avg = input.nextDouble();
```

```
        char ch;
```

```
        if(avg >= 75) {
```

```
            ch = 'A'; // 75 - 100
```

```
        } else if(avg >= 65){
```

```
            ch = 'B'; // 65 - 75
```

```
        } else if(avg >=50){
```

```
            ch = 'C'; // 50 - 65
```

```
        } else {
```

```
            ch = 'F'; // <50
```

```
        }
```

```
        System.out.println("Grade: " + ch);
```

```
    }
```

```
}
```

////////////////////////////////////

Q190 From Q189

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input marks: ");
        double avg = input.nextDouble();
        char ch;
        if(avg >= 75) {
            ch = 'A'; // 75 - 100
        } else
            if(avg >= 65){
                ch = 'B'; // 65 - 75
            } else
                if(avg >=50){
                    ch = 'C'; // 50 - 65
                } else {
                    ch = 'F'; // <50
                }

        ch = avg >= 75 ? 'A' : (avg>=65 ? 'B' :
        (avg>=50 ? 'C' : 'F'));

        System.out.println("Grade: " + ch);
    }
}
```



```

char ch;
if (avg >= 75) {
    ch = 'A'; // 75 - 100
} else
    if (avg >= 65) {
        ch = 'B'; // 65 - 75
    } else
        if (avg >= 50) {
            ch = 'C'; // 50 - 65
        } else {
            ch = 'F'; // <50
        }
    }
}

```

$ch = avg \geq 75 ? 'A' : avg \geq 65 ? 'B' : avg \geq 50 ? 'C' : 'F';$

//////////////////////////////////////

Q191

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input marks: ");
```

```
        double avg = input.nextDouble();
```

```
        char ch;
```

```
        if (avg >= 75) {
```

```
            ch = 'A'; // 75 - 100
```

```
        } else
```

```
            if (avg >= 65){
```

```
                ch = 'B'; // 65 - 75
```

```
            } else
```

```
                if (avg >= 50){
```

```
                    ch = 'C'; // 50 - 65
```

```
                } else {
```

```
                    ch = 'F'; // <50
```

```
                }
```

```
        ch =
```

```
            avg >= 75 ? 'A' :
```

```
            avg >= 65 ? 'B' :
```

```
            avg >= 50 ? 'C' : 'F';
```

```

    }
}
////////////////////////////////////

```

}

////////////////////////////////////

////////////////////////////////////

```
import java.util.*;
```

```
public static void main(String args[]){
```

```
System.out.print("Input x: ");
```

```
int x = input.nextInt();
```

```
if(x == 100) {
```

```
System.out.println("A");
```

```
} else if(x == 200){
```

```
System.out.println("B");
```

```
} else if(x == 300){
```

```
System.out.println("C");
```

```
} else if(x == 400){
```

```
System.out.println("D");
```

```
} else {
```

```
System.out.println("E");
```

}

}

}

////////////////////////////////////

Q194 From Q193

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input x: ");
        int x = input.nextInt();
        /*
        if(x == 100) {
            System.out.println("A");
        } else if(x == 200){
            System.out.println("B");
        } else if(x == 300){
            System.out.println("C");
        } else if(x == 400){
            System.out.println("D");
        } else {
            System.out.println("E");
        }
        */
        switch(x){
            case 100: System.out.println("A");
            case 200: System.out.println("B");
            case 300: System.out.println("C");
            case 400: System.out.println("D");
            default: System.out.println("E");
        }
    }
}
```

```
Input x: 100
A
B
C
D
E

-----
(program exited with code: 0)
```

Q195

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input x: ");
        int x = input.nextInt();

        switch(x){
            case 100:
                System.out.println("A");
            case 200:
                System.out.println("B");
            case 300:
                System.out.println("C");
            case 400:
                System.out.println("D");
            default:
                System.out.println("E");
        }
    }
}
```

////////////////////////////////////
Q196

import java.util.*;

class Example{

public static void main(String args[]){

Scanner input = new Scanner(System.in);

System.out.print("Input x: ");

int x = input.nextInt();

switch(x){

case 100:

System.out.println("A");

break;

case 200:

System.out.println("B");

break;

case 300:

System.out.println("C");

break;

case 400:

System.out.println("D");

break;

default:

System.out.println("E");

break;

}

}

}

////////////////////////////////////
Q197

import java.util.*;

class Example{

public static void main(String args[]){

Scanner input = new Scanner(System.in);

System.out.print("Input x: ");

int x = input.nextInt();

switch(x){

case 100:

System.out.println("A");

break;

case 200:

System.out.println("B");

break;

case 300:

System.out.println("C");

break;

case 400:

System.out.println("D");

break;

default:

System.out.println("E");

// break;

}

}

}

////////////////////////////////////

Q198 Exercise - Using "if" statement

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input month: ");
        int month = input.nextInt();

        System.out.println("31 days");
        System.out.println("30 days");
        System.out.println("28 days");
    }
}
```


////////////////////////////////////

Q199 From Q198

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input month: ");
        int month = input.nextInt();

        if(month == 1 || month == 3 || month == 5 ||
month == 7
                                || month == 8 || month == 10
|| month == 12) {
            System.out.println("31 days");
        } else if (month == 4 || month == 6 || month ==
9 || month == 11) {
            System.out.println("30 days");
        } else if (month == 2) {
            System.out.println("28 days");
        } else {
            System.out.println("Invalid Month");
        }
    }
}
```

////////////////////////////////////

Q200 From Q199 : Using "Switch/Case"

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input month: ");
        int month = input.nextInt();

        switch(month) {
            case 1:
            case 3:
            case 5:
            case 7:
            case 8:
            case 10:
            case 12:
                System.out.println("31 days");
                break;
            case 4:
            case 6:
            case 9:
            case 11:
                System.out.println("30 days");
                break;
            case 2:
                System.out.println("28 days");
                break;
            default:
                System.out.println("Invalid month");
        }
    }
}
```

////////////////////////////////////

Q201

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        char ch = 'A';
```

```
        switch(ch) {
```

```
            case 'A':
```

```
                System.out.println("---");
```

```
                break;
```

```
            case 'B':
```

```
                System.out.println(+++);
```

```
        }
```

```
    }
```

```
}
```

////////////////////////////////////

Q202 Case I

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){

        /* insert code */
        switch(x) {

        }

    }
}
```

//byte -

//short -

//int -

//long - Illegal

//float - Illegal

//double - Illegal

//char

//boolean - Illegal

//String

//enum

```
////////////////////////////////////
```

Q203 Case II

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 100;

        switch(x) {

        }

    }
}
```

```
////////////////////////////////////
```

Q204 Case III

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 100;

        switch(x) {
            case 100:
                System.out.println("---");
                break;
            case 200:
                System.out.println(+++);
                break;
        }
    }
}
```

////////////////////////////////////

Q205 Case IV

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 100;

        switch(x) {
            case 100:
                System.out.println("---");
                break;
            default:
                System.out.println("default");
                break;
            case 200:
                System.out.println(+++);
                break;
        }
    }
}
```

////////////////////////////////////
Q206 Case V

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 100;

        switch(x) {
            case 100:
                System.out.println("---");
                break;
            case 200:
                System.out.println(+++);
                break;
            case 100: // Illegal
                System.out.println("--- 2");
                break;
            default:
                System.out.println("default");
                break;
        }
    }
}
```

////////////////////////////////////

Q207 Case VI

=====

```
class Example{
    public static void main(String args[]){
        int x = 100;

        switch(x) {
            case 100:
                System.out.println("---");
                break;
            case 'A':
                System.out.println(+++);
                break;
            default:
                System.out.println("default");
                break;
        }
    }
}
```


////////////////////////////////////

Q208 Case VII

=====

```
class Example{
    public static void main(String args[]){
        byte x = 100;

        switch(x) {
            case 100:
                System.out.println("---");
                break;
            case 200: // Illegal
                System.out.println(+++);
                break;
            default:
                System.out.println("default");
                break;
        }
    }
}
```

////////////////////////////////////
Q209 Case VIII

=====

```
class Example{
    public static void main(String args[]){
        int x = 100;

        int a = 100, b = 200, c = 300;
        switch(x) {
            case a:
                System.out.println("---a");
                break;
            case b:
                System.out.println(++b);
                break;
            case c:
                System.out.println(++c);
                break;
            default:
                System.out.println("default");
                break;
        }
    }
}
```

////////////////////////////////////

Q210 Final Variables - Constant

=====

```
class Example{
    public static void main(String args[]){
        final int x = 100;

        x = 200; // Illegal
    }
}
```

////////////////////////////////////
Q211 Final Variables - Constant

=====

```
class Example{
    public static void main(String args[]){
        final int x;
        x = 100; // Legal
        x = 200; // Illegal
    }
}
```

////////////////////////////////////

Q212 Case IX

=====

```
class Example{
    public static void main(String args[]){
        int x = 100;

        final int a = 100, b = 200, c = 300;
        switch(x) {
            case a:
                System.out.println("---a");
                break;
            case b:
                System.out.println(++b);
                break;
            case c:
                System.out.println(++c);
                break;
            default:
                System.out.println("default");
                break;
        }
    }
}
```

Q213

Variable Scope and Lifetime

Q214

```
class Example{
    public static void main(String args[]){
        int x = 100;

        int x = 200; // Illegal
    }
}
```

Q215

```
class Example{
    public static void main(String args[]){
        int x = 100;

        {
            int y = 200;

            System.out.println(y);
        }
        System.out.println(y); // Illegal
        //int x = 200;
    }
}
```

//

Q216

```
class Example{
    public static void main(String args[]){
        int x = 100;

        {
            int y = 200;

            System.out.println(y);
        }
        int y = 100;
        //int x = 200;
    }
}
```

//

Q217

```
class Example{
    public static void main(String args[]){
        int x = 100;

        if(x > 100){
            int y = 200;
            System.out.println(y);
        }
        int y = 100;
        //int x = 200;
    }
}
```

//

Q218

```
class Example{
    public static void main(String args[]){
        int x = 100;
        int y = 100;
        {
            System.out.println(y);
            int y = 200; // Illegal
            System.out.println(y);
        }
    }
}
```

////////////////////////////////////
Q219

```
class Example{  
    public static void main(String args[]){  
        int x = 100;  
  
        {  
            int y = 200;  
            System.out.println(y);  
        }  
  
        {  
            int y = 100;  
            System.out.println(y);  
        }  
    }  
}
```

////////////////////////////////////

Variable Initialization

////////////////////////////////////

Q220

```
class Example{  
    public static void main(String args[]){  
        int x;  
  
        {  
            x = 200;  
        }  
  
        System.out.println(x);  
    }  
}
```

////////////////////////////////////
Q221

```
class Example{  
    public static void main(String args[]){  
        int x;  
  
        if(true){  
            x = 200;  
        }  
  
        System.out.println(x);  
    }  
}
```

////////////////////////////////////
Q222

```
class Example{  
    public static void main(String args[]){  
        int x;  
        boolean b = true;  
        if(b){  
            x = 200;  
        }  
  
        System.out.println(x);  
    }  
}
```


Q223

```
class Example{
    public static void main(String args[]){
        int x;
        final boolean b = true;
        if(b){
            x = 200;
        }

        System.out.println(x);
    }
}
```

Java Flow Control "for loop"

Q224

[illegible]

//

Q225

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<100; i++) System.out.println("A");
    }
}
```

//

Q226

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<100; i++)
            System.out.println("A");
        System.out.println("B");
    }
}
```

//

Q227

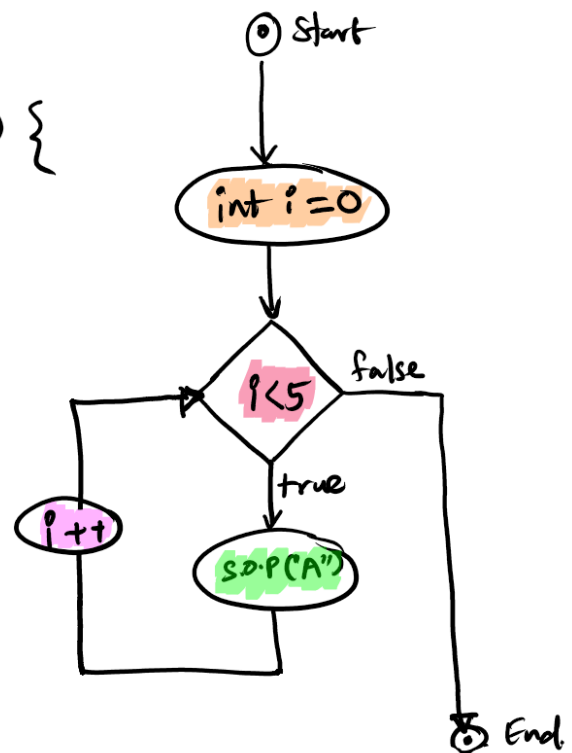
```
class Example{
    public static void main(String args[]){
        for(int i=0; i<100; i++)
            System.out.println("A");
        System.out.println("B");
    }
}
```

Q228

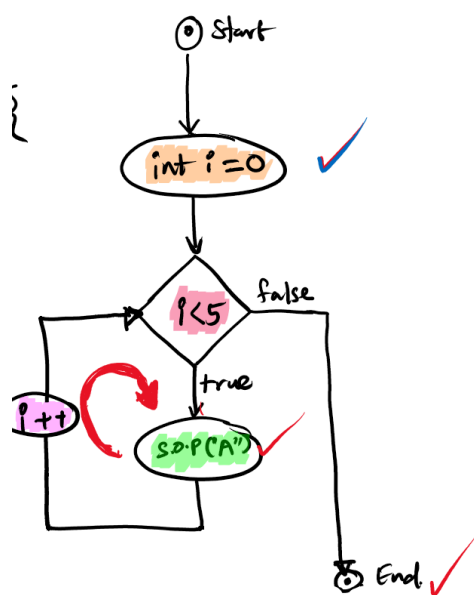
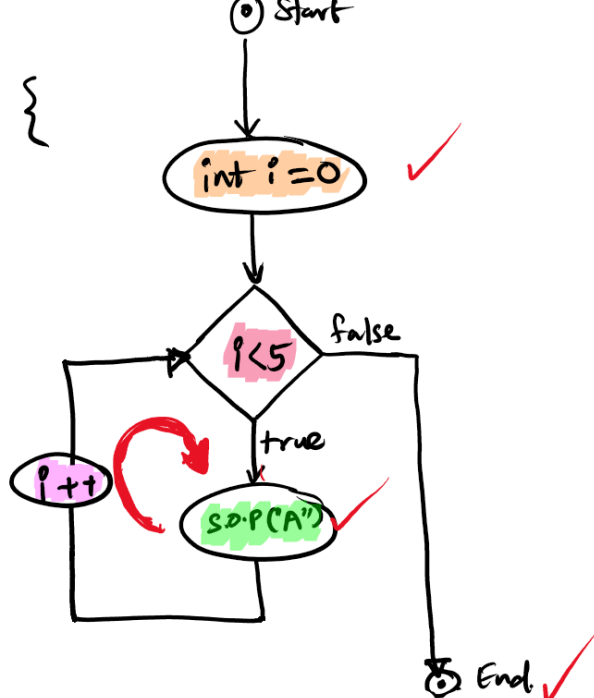
```
class Example{
    public static void main(String args[]){
        for(int i=0; i<100; i++){
            System.out.println("A");
            System.out.println("B");
        }
    }
}
```

Q229

①
for (int i=0; i<5; i++) {
 S.O.P("A")
}
}



```
for (int i=0; i<5; i++) {
    S0-P("A")
}
```



i → 0 1 2 3 4 5

0 < 5 ✓
 1 < 5 ✓
 2 < 5 ✓
 3 < 5 ✓
 4 < 5 ✓
 5 < 5 ✗

A ✓
 A ✓
 A ✓
 A ✓
 A ✓

//

Q230 Case I

=====

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5; i++){
            System.out.println(i);
        }
    }
}
```

//

Q231 Case II

=====

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5; i++){
            System.out.println(i);
        }

        System.out.println(i);
    }
}
```

//

Q232 Case III

=====

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<=5; i++){
            System.out.println(i);
        }
    }
}
```

Q233 Case IV

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<=5; i+=2){
            System.out.println(i);
        }
    }
}
```

Q234 Case V

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<=5;){
            System.out.println(i);
        }
    }
}
```

Q235

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<=5;){
            System.out.println(i++);
        }
    }
}
```

//

Q236

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<=5;){
            System.out.println(++i);
        }
    }
}
```

//

Q237

```
class Example{
    public static void main(String args[]){
        int i = 0;
        for(;i<=5;){
            System.out.println(i);
        }
    }
}
```

//

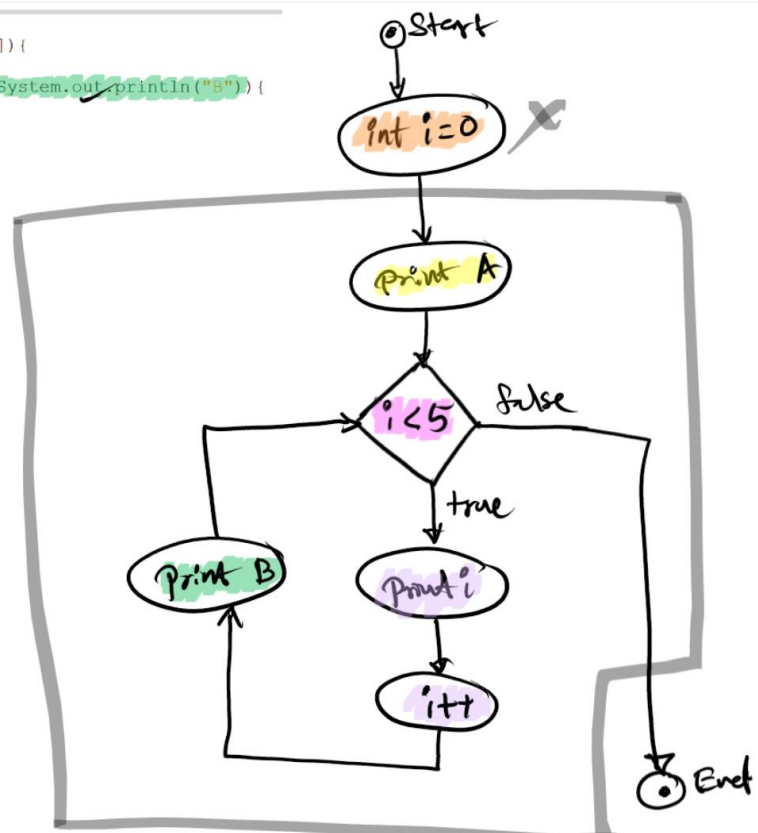
Q238

```
class Example{
    public static void main(String args[]){
        int i = 0;
        for(;i<5;){
            System.out.println(i);
            i++;
        }
    }
}
```

Q239

```
class Example{  
    public static void main(String args[]){  
        int i = 0;  
  
        for(System.out.println("A");i<5;System.out.println("B  
")){  
            System.out.println(i);  
            i++;  
        }  
    }  
}
```

```
class Example{  
    public static void main(String args[]){  
        int i = 0;  
        for (System.out.println("A"); i < 5; System.out.println("B")) {  
            System.out.println(i);  
            i++;  
        }  
    }  
}
```




```

        System.out.println("Avg: " +
(double)total/numOfSubjects);
    }
}

```

```

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

```

Q244 From 242

```

=====

```

```

import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter num1: ");
        int num1 = input.nextInt();
        System.out.print("Enter num2: ");
        int num2 = input.nextInt();

        System.out.print("[");

        if(num1 < num2){
            for(int i=num1; i<=num2; i++){
                System.out.print(i+", ");
            }
        } else {
            for(int i=num2; i<=num1; i++){
                System.out.print(i+", ");
            }
        }
        System.out.print("\b\b]");
    }
}

```

////////////////////////////////////
Q245

```
class Example{  
    public static void main(String args[]){  
        for(int i=0; i<10; i++){  
            if(i%2 == 0) {  
                System.out.println(i);  
            }  
        }  
    }  
}
```

////////////////////////////////////

Continue

////////////////////////////////////

Q246

```
class Example{  
    public static void main(String args[]){  
        for(int i=0; i<10; i++){  
  
            if(i%2 == 1){  
                continue;  
            }  
  
            System.out.println(i);  
        }  
    }  
}
```

```
graph TD
    Start((Start)) --> Init([int i = 0])
    Init --> Cond1{i < 10}
    Cond1 -- false --> End((End))
    Cond1 -- true --> Cond2{i % 2 == 1}
    Cond2 -- true --> Inc((i++))
    Inc --> Cond1
    Cond2 -- false --> Print([Print i])
    Print --> End
```

Q247

```
if(i%2 == 1){
    continue;
    System.out.println("A"); //Illegal
}
```

```
System.out.println(i);
```

}

}

}

```
////////////////////////////////////
Q248
import java.util.*;
class Example{
    public static void main(String args[]){
        for(int i=0; i<10; i++){
            continue;
            System.out.println(i); // Illegal
        }
    }
}
////////////////////////////////////
```

Break

```
////////////////////////////////////
Q249
class Example{
    public static void main(String args[]){
        for(int i=0;;i++){
            if(i>=5){
                break;
            }
            System.out.println("A");
        }
    }
}
////////////////////////////////////
```

```
Q250
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter no of subjects: ");
        int numofSubjects = input.nextInt();
    }
}
```

```

        int total = 0;
        for(int i=0; i<numOfSubjects; i++){
            System.out.print("Enter marks for
subject"+(i+1)+" ": );
            int marks = input.nextInt();
            if(marks<50) {
                break;
            }
            total += marks;
        }

        System.out.println("Total: " + total);
        System.out.println("Avg: " +
(double)total/numOfSubjects);
    }
}

```

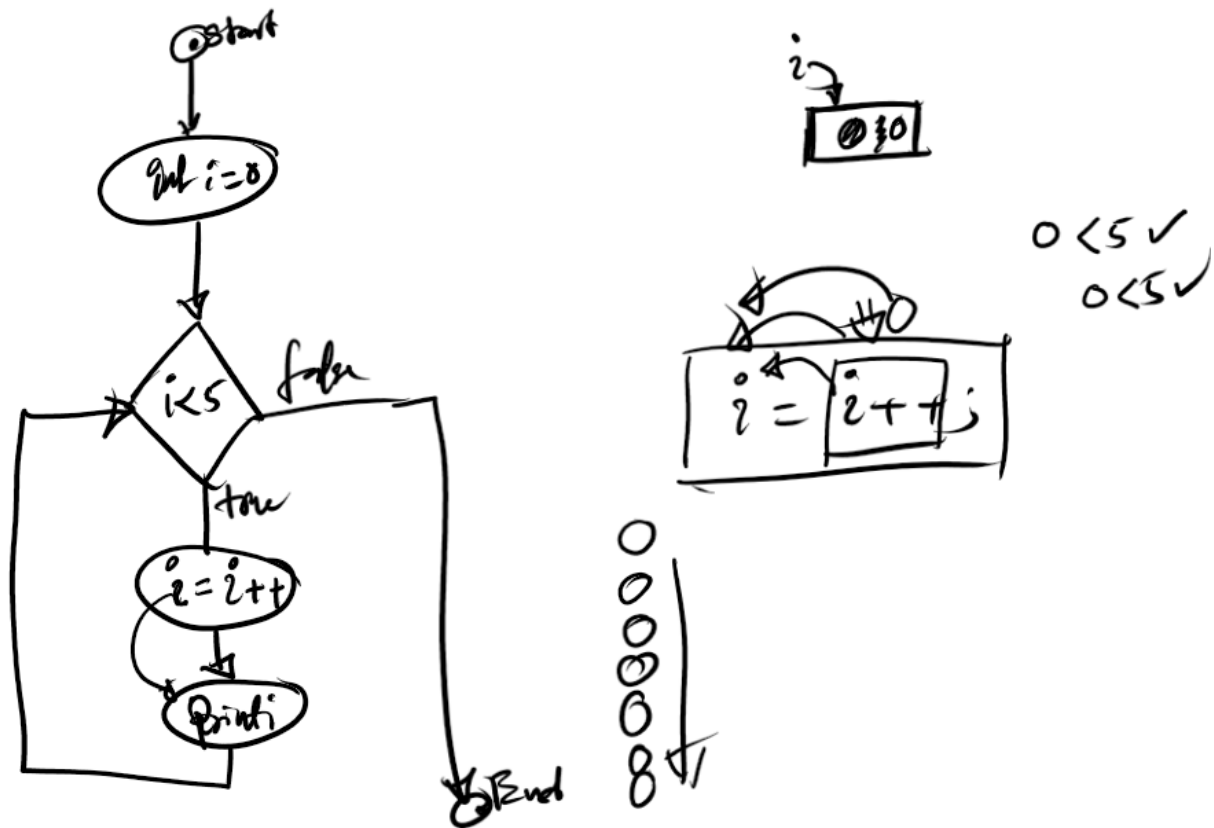
//

Q251

```

import java.util.*;
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;){
            System.out.println(i=i++);
        }
    }
}

```



}

////////////////////////////////////
Q253

```
import java.util.*;
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            if(i > 3) {
                return;
            }
            System.out.println(i);
        }

        System.out.println("End main()");
    }
}
```

////////////////////////////////////

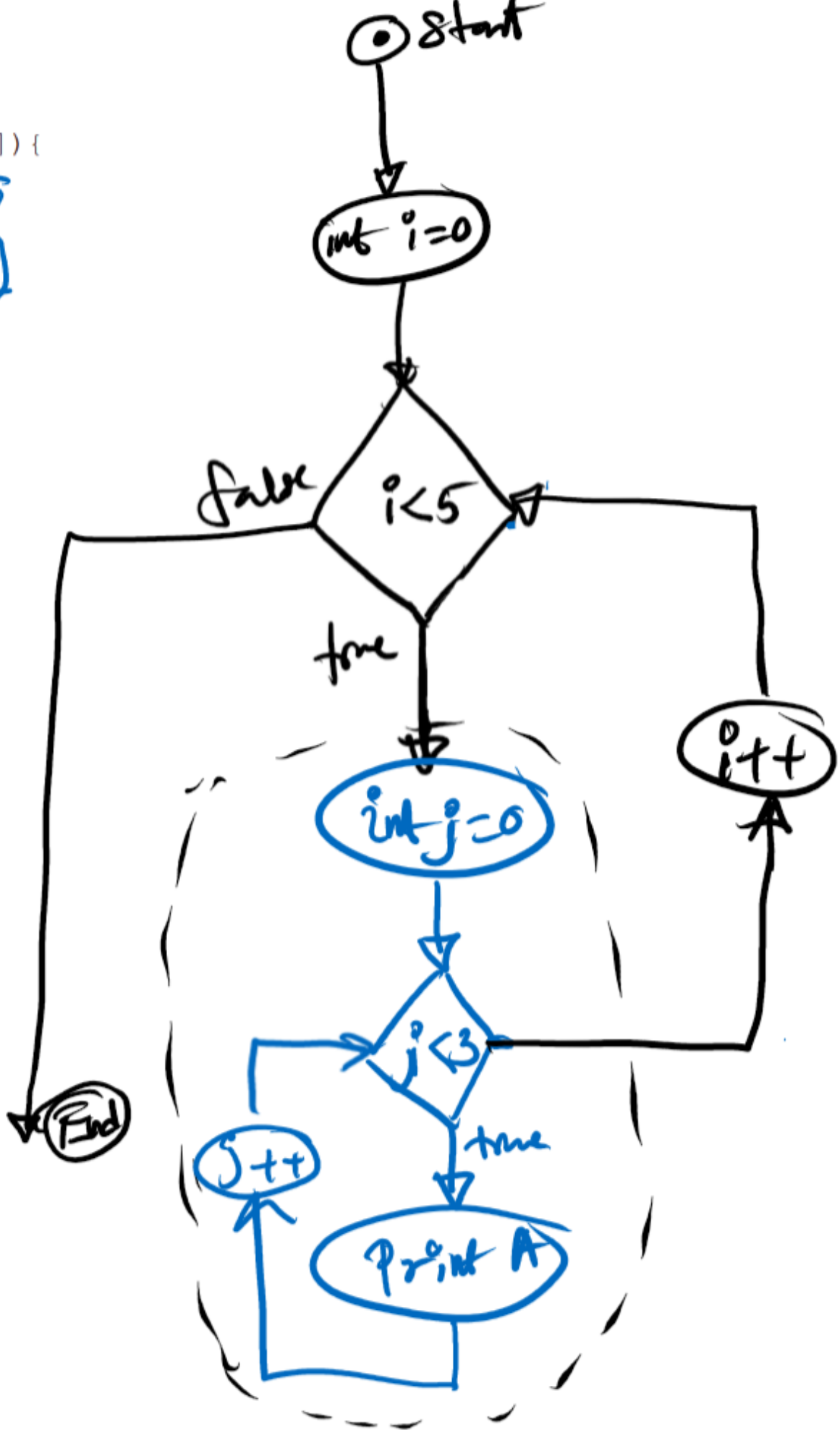
Nested Loops

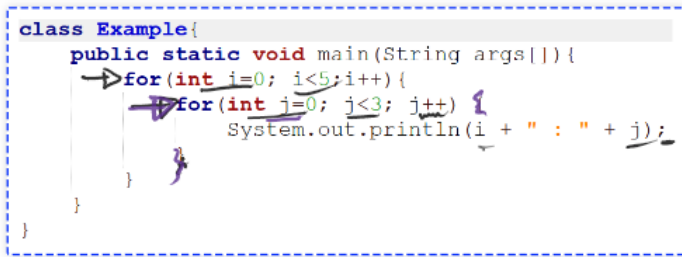
////////////////////////////////////

Q254

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<3; j++) {
                System.out.println("A");
            }
        }
    }
}
```

```
[[ ] {  
  ;  
}]
```



 $0 < 3$ ✓ $1 < 3 \checkmark$ $2 < 3$ ✓

323/

 $1 < 5 \checkmark$
$$0:0$$

✓ O : 1

~~0~~:2

 $i:0$

131

 $1:2$

2:0

2:1

232

3:00

3:1

3:2

4:0

421

4:2

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<3; j++) {
                System.out.println(i + " : " + j);
            }
        }
    }
}
```

Q255

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<3; j++) {
```

```

        for(int k=0; k<2; k++) {
            }
        }
    }
}
////////////////////////////////////

```

Q256

```

class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=i; j<3; j++) {
                System.out.println(i + " : " + j);
            }
        }
    }
}

```



```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=i; j<3; j++) {
                System.out.println(i + " : " + j);
            }
        }
    }
}
```

0	1
0	2
1	1
1	2
2	2

Q257

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<i; j++) {
                System.out.println(i + " : " + j);
            }
        }
    }
}
```

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<i; j++) {
                System.out.println(i + " : " + j);
            }
        }
    }
}
```

1:0 4:0
2:0 4:1
2:1 4:2
3:0 4:3
3:1
3:2

?
✓ 0 → ~~0~~ 0 j < 1
✓ 1 → ~~0~~ 1 j < 2
✓ 2 → j=0 j < 3
✓ 3 → j=0 j < 4
 0, 1, 2
✓ 4 → j=0 j < 5
 0, 1, 2, 3

//

Q258

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<i; j++) {

            }
            System.out.println(i + " : " + j);
        }
    }
}
```

//

Q259

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<i; j++) {

            }
            System.out.println(i + " : " + j); // j - compile
error
        }
    }
}
```

//

Q260

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<10;i++){
            for(int j=0; j<10; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

```

    }
}
}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

```

Q261

```

class Example{
    public static void main(String args[]){
        for(int i=0; i<10;i++){
            for(int j=0; j<i; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
}

```

```

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

```

Q262

```

class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<i; j++) {

                if(i>3){
                    break;
                }

                System.out.println("Inner Loop");
            }
            System.out.println("Outer Loop");
        }
    }
}
}

```



```

class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<i; j++) {

                if(i>3){
                    break;
                }

                System.out.println("Inner Loop");
            }
            System.out.println("Outer Loop");
        }
    }
}

```

0 → X Outer Loop
 1 → j=0 j<1 Inner Loop
 2 → j=0 j<2 Inner Loop
 3 → j=0 j<3 Inner Loop
 4 → j=0 j<4 Outer Loop

////////////////////////////////////

Q263

```

class Example{
    public static void main(String args[]){
        L1:for(int i=0; i<5;i++){
            L2:for(int j=0; j<i; j++) {

                if(i>3){
                    break L1;
                }

                System.out.println("Inner Loop");
            }
            System.out.println("Outer Loop");
        }

        System.out.println("End of main()");
    }
}

```

////////////////////////////////////
Q264

```
class Example{  
    public static void main(String args[]){  
        for(int i=5; i>0;i--){  
            System.out.println(i);  
        }  
    }  
}
```

////////////////////////////////////
Q265 Exercise

=====

```
import java.util.*;  
class Example{  
    public static void main(String args[]){  
        Scanner input = new Scanner(System.in);  
  
        int total = 0;  
  
        System.out.println("Total: " + total);  
        System.out.println("Avg: " + );  
    }  
}
```

////////////////////////////////////
Q266

import java.util.*;

class Example{

public static void main(String args[]){

Scanner input = new Scanner(System.in);

int total = 0;

for(;;){

System.out.print("Enter number: ");

int num = input.nextInt();

if(num == -1) {

break;

}

total += num;

}

System.out.println("Total: " + total);

//System.out.println("Avg: " +);

}

}

//

Q267

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);

        int total = 0;
        int num = 0;
        for(;num != -1;){
            System.out.print("Enter number: ");
            num = input.nextInt();

            total += num;
        }

        System.out.println("Total: " + total);
        //System.out.println("Avg: " + );
    }
}
```

//

While Loop

//

Q268

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);

        int total = 1;
        int num = 0;
        while(num != -1){ // => for(;num!=-1;){
            System.out.print("Enter number: ");
            num = input.nextInt();
        }
    }
}
```

```

        total += num;
    }

    //total++;

    System.out.println("Total: " + total);
    //System.out.println("Avg: " + );
}
}
////////////////////////////////////

```

Q269

```

import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);

        int total = 1;
        int num;

        System.out.print("Enter number: ");
        num = input.nextInt();

        total += num;

        while(num != -1){ // => for(;num!=-1;){
            System.out.print("Enter number: ");
            num = input.nextInt();

            total += num;
        }
        System.out.println("Total: " + total);
    }
}

```

do/while loop

Q270

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        int total = 1;
        int num;
        do{
            System.out.print("Enter number: ");
            num = input.nextInt();

            total += num;
        }while(num != -1);
        System.out.println("Total: " + total);
    }
}
```

Q271

```
class Example{
    public static void main(String args[]){
        for(;;){

        } // legal
        while(){ // Illegal

        }
        do{

        }while(); // Illegal
    }
}
```

Q272 Exercise I

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.println("Enter 1 for DecimalToBinary");
        System.out.println("Enter 2 for DecimalToOctal");
        System.out.println("Enter 3 for DecimalToHexa");
        System.out.println("Enter 99 to Exit");

        System.out.print("Select the converter: ");

        System.out.println("DecimalToBinary Converter"); // 1
        System.out.println("Enter 0 to Back");
        System.out.println("Enter 99 to Exit");
        System.out.print("Select the action: ");

        System.out.println("DecimalToOctal Converter"); // 2
        System.out.println("Enter 0 to Back");
        System.out.println("Enter 99 to Exit");
        System.out.print("Select the action: ");

        System.out.println("DecimalToHexa Converter"); // 3
        System.out.println("Enter 0 to Back");
        System.out.println("Enter 99 to Exit");
        System.out.print("Select the action: ");

        System.out.println("Invalid Input...");

    }
}
```

////////////////////////////////////

Q273 From Q272

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);

        while(true) {
            System.out.println("Enter 1 for DecimalToBinary");
            System.out.println("Enter 2 for DecimalToOctal");
            System.out.println("Enter 3 for DecimalToHexa");
            System.out.println("Enter 99 to Exit");

            System.out.print("Select the converter: ");
            int option = input.nextInt();

            if(option == 1){
                System.out.println("DecimalToBinary Converter");
            } else if (option == 2){
                System.out.println("DecimalToOctal Converter");
            } else if (option == 3) {
                System.out.println("DecimalToHexa Converter");
            } else if (option == 99) {
                return;
                //System.exit(0);
            } else {
                System.out.println("Invalid Input...");
            }
        }
    }
}
```



```
while(true){
```

```

--");

Converter");

--");

");

int action = input.nextInt();

if(action == 0) {
    continue L1;
} else if (action == 99) {
    return;
} else {
    System.out.println("Invalid
Input...");
}
case 3:
while(true){
    System.out.println("-----

--");

Converter");

System.out.println("-----

--");

System.out.println("Enter 0 to Back");
System.out.println("Enter 99 to Exit");
System.out.print("Select the action:

");

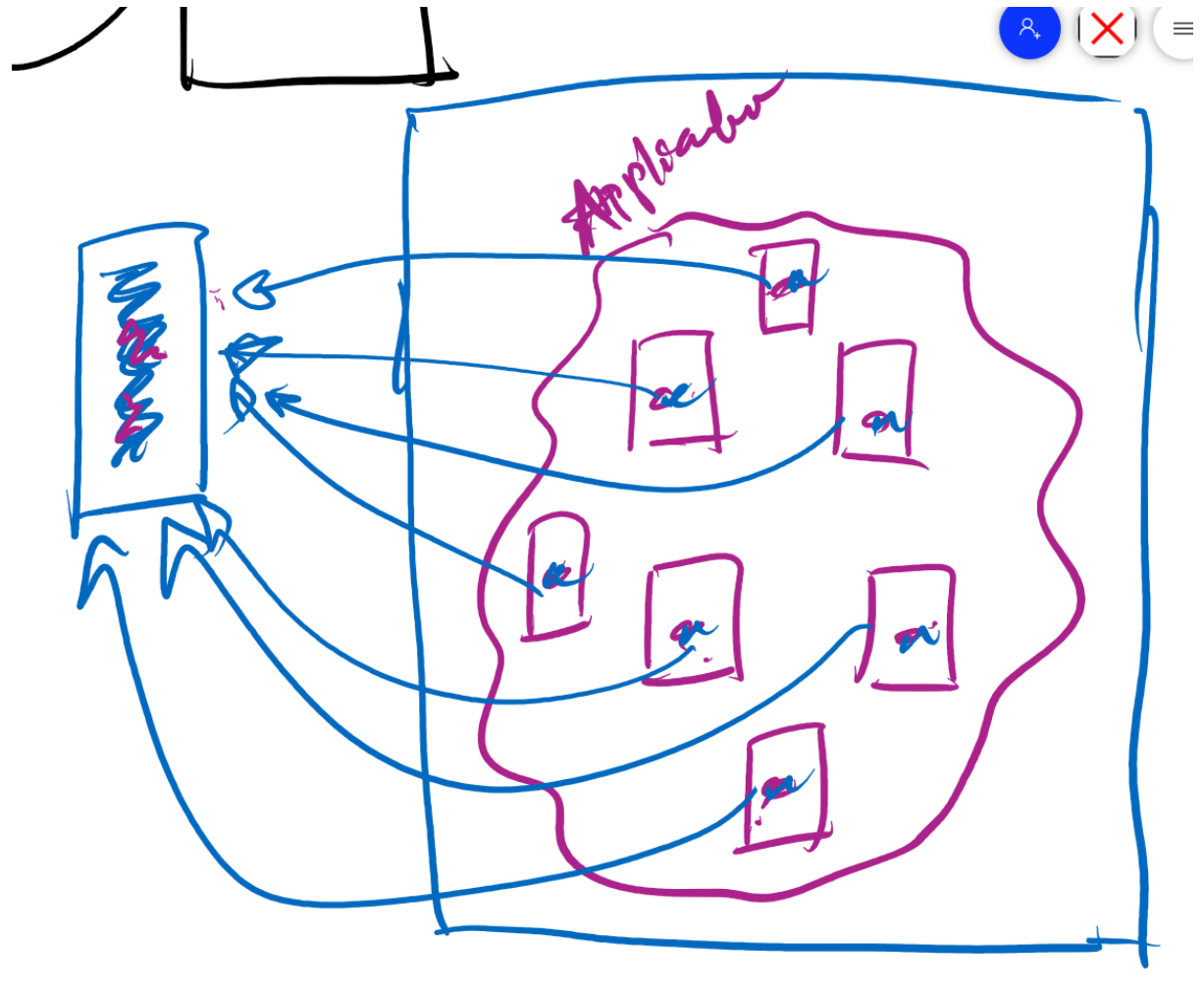
int action = input.nextInt();

if(action == 0) {
    continue L1;
} else if (action == 99) {
    return;
} else {
    System.out.println("Invalid
Input...");
}
}

```

```
case 99:  
    return;  
default:  
    System.out.println("Invalid Input...");  
}  
}  
}  
}
```

////////////////////////////////////
Q275



Java Methods

Q276

```
import java.util.*;
class Example{
    public static void main(String args[]){

        System.out.println("Dinuka Kodithuwakku");
        System.out.println("No 23/5");
        System.out.println("Panadura Rd");
        System.out.println("Panadura");

        //
        //
        //

        System.out.println("Dinuka Kodithuwakku");
        System.out.println("No 23/5");
        System.out.println("Panadura Rd");
        System.out.println("Panadura");

        //
        //
        //
        //

        System.out.println("Dinuka Kodithuwakku");
        System.out.println("No 23/5");
        System.out.println("Panadura Rd");
        System.out.println("Panadura");
        //
        //
        //
        //
        //
        System.out.println("Dinuka Kodithuwakku");
        System.out.println("No 23/5");
        System.out.println("Panadura Rd");
        System.out.println("Panadura");

    }
}
```

}

`public static void printTotal() {`
// Method Body / Implementation
`}`

method name (pointing to `printTotal()`)

```

import java.util.*;
class Example{

    public static void printTotal(){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter num1: ");
        int num1 = input.nextInt();

        System.out.print("Enter num2: ");
        int num2 = input.nextInt();

        int total = num1 + num2;

        System.out.println("Total of "+num1+" and "+num2+" => " +
total);
    }

    public static void main(String args[]){
        printTotal();
    }
}
  
```

}

Parameterized Methods

Q281

```
import java.util.*;
```

```
class Example{
```

```
    public static void printTotal(int num1, int num2){
```

```
        // ----- Process -----
```

```
        int total = num1 + num2;
```

```
        // ----- Output -----
```

```
        System.out.println("Total of "+num1+" and " +num2+" => " +
```

```
total);
```

```
    }
```

```
    public static void main(String args[]){
```

```
        // ----- Input -----
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Enter num1: ");
```

```
        int num1 = input.nextInt();
```

```
        System.out.print("Enter num2: ");
```

```
        int num2 = input.nextInt();
```

```
        printTotal(num1, num2);
```

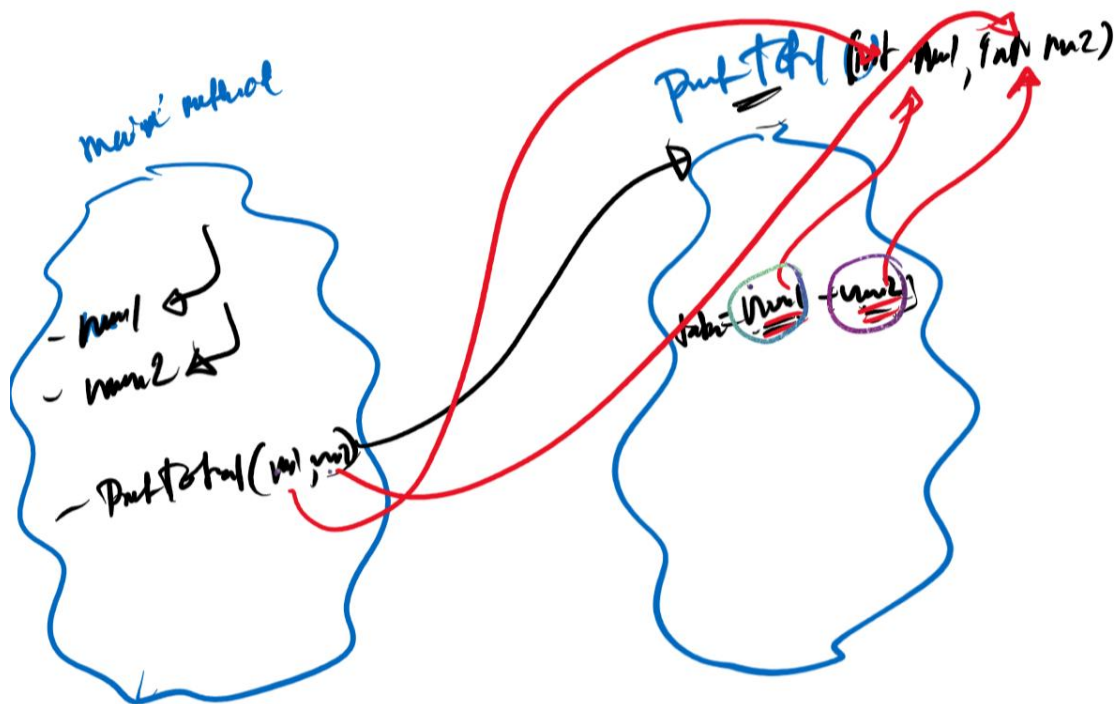
```
    }
```

```
}
```

Handwritten notes and annotations:

- Parameters list* (green text) with an arrow pointing to the parameters `(int num1, int num2)` in the `printTotal` method signature.
- Public static void printTotal (int num1, int num2) {* (handwritten in black ink, with the method signature highlighted in pink).
- // ----- Process -----* (handwritten in black ink, with the comment highlighted in grey).
- int total = num1 + num2;* (handwritten in black ink, with the line highlighted in grey).
- // ----- Output -----* (handwritten in black ink, with the comment highlighted in grey).
- System.out.println("Total of "+num1+" and " +num2+" => " +* (handwritten in black ink, with the line highlighted in grey).
- total);* (handwritten in black ink, with the line highlighted in grey).
- }* (handwritten in black ink, with the closing brace highlighted in grey).
- Public static void main (String args[]) {* (handwritten in black ink, with the method signature highlighted in pink).
- //* (handwritten in black ink, with the comment highlighted in pink).
- printTotal (num1, num2);* (handwritten in black ink, with the method call highlighted in pink).
- arguments* (green text) with an arrow pointing to the arguments `num1, num2` in the `printTotal` method call.
- }* (handwritten in black ink, with the closing brace highlighted in pink).

Q282



Q283

```
import java.util.*;
class Example{

    public static void printTotal(int a, int b){
        // ----- Process -----
        int total = num1 + num2;

        // ----- Output -----
        System.out.println("Total of "+num1+" and " +num2+" => " +
total);
    }

    public static void main(String args[]){
        // ----- Input -----
        Scanner input = new Scanner(System.in);
        System.out.print("Enter num1: ");
        int num1 = input.nextInt();
        System.out.print("Enter num2: ");
        int num2 = input.nextInt();

        printTotal(num1, num2);
    }
}
```

Q284

```
import java.util.*;
```

```
class Example{
```

```
    public static void printTotal(int a, int b){
```

```
        // ----- Process -----
```

```
        int total = a + b;
```

```
        // ----- Output -----
```

```
        System.out.println("Total of "+a+" and "+b+" => "+ total);
```

```
    }
```

```
    public static void main(String args[]){
```

```
        // ----- Input -----
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Enter num1: ");
```

```
        int num1 = input.nextInt();
```

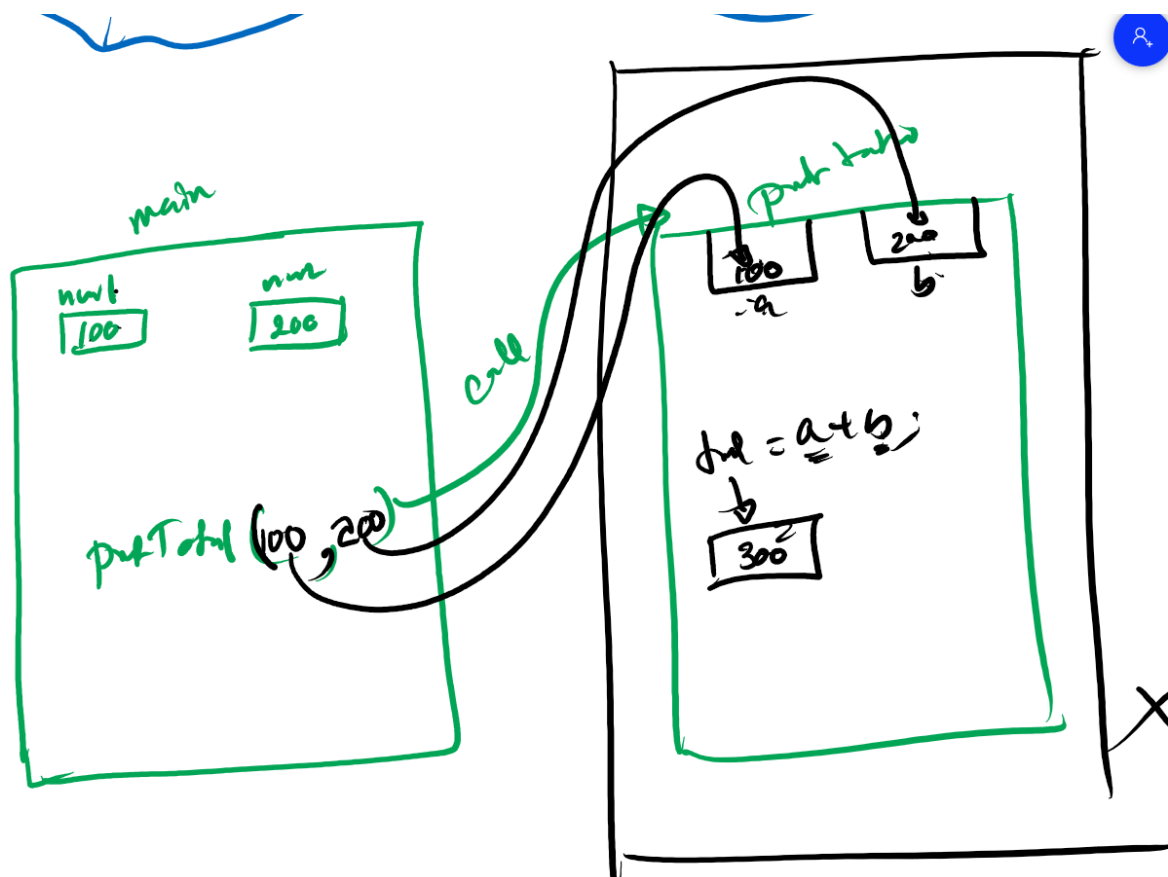
```
        System.out.print("Enter num2: ");
```

```
        int num2 = input.nextInt();
```

```
        printTotal(num1, num2);
```

```
    }
```

```
}
```



}

}

```
// 0! = 1;
```

```
Scanner input = new Scanner(System.in);
```

```

        System.out.print("Enter a number: ");
        int num = input.nextInt();

        System.out.println("Reversed value of " + num + " => " +
reverse(num));
    }
}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

```

Q293

```

import java.util.*;
class Example{

    public static boolean isPalindrom(int number) {
        if(number == reverse(number)) {
            return true;
        } else {
            return false;
        }
    }

    public static int reverse(int number) {
        int reversedNumber = 0;
        while(number != 0){
            reversedNumber = reversedNumber * 10 + (number %
10);

            number /= 10;
        }

        return reversedNumber;
    }

    public static void main(String args[]){
        // ----- Input -----
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = input.nextInt();

        System.out.println("Reversed value of " + num + " => " +
reverse(num));
    }
}

```

```
int reversedNumber = 0;
```



```

        while(number != 0){
            reversedNumber = reversedNumber * 10 + (number %
10);

            number /= 10;
        }

        return reversedNumber;
    }

    public static void main(String args[]){
        // ----- Input -----
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = input.nextInt();
        System.out.println("Reversed value of " + num + " => " +
reverse(num));
        System.out.println("Is Palindrom: " + isPalindrom(num));
    }
}

```

//

Method Calling Stack

//

Q296

```

class Example{
    public static void methodA() {
        System.out.println("Start methodA()");
        methodB();
        System.out.println("End methodA()");
    }

    public static void methodB() {
        System.out.println("Start methodB()");

        System.out.println("End methodB()");
    }

    public static void main(String args[]){
        System.out.println("Start main()");

        methodA();

        System.out.println("End main()");
    }
}

```

Q297

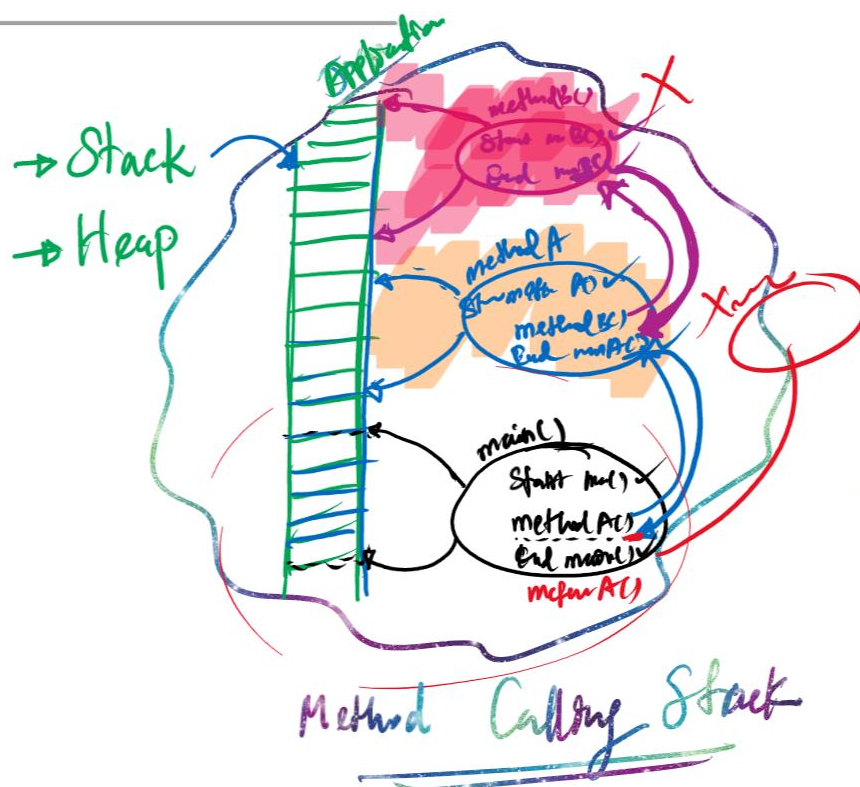
```

1 class Example{
2     public static void methodA() {
3         System.out.println("Start methodA()"); ✓
4         methodB(); ✓
5         System.out.println("End methodA()"); ✓
6     }
7
8     public static void methodB() {
9         System.out.println("Start methodB()"); ✓
10        System.out.println("End methodB()"); ✓
11    }
12 }
13
14 public static void main(String args[]){
15     System.out.println("Start main()"); ✓
16     methodA(); ✓
17     System.out.println("End main()"); ✓
18 }
19
20 }
21

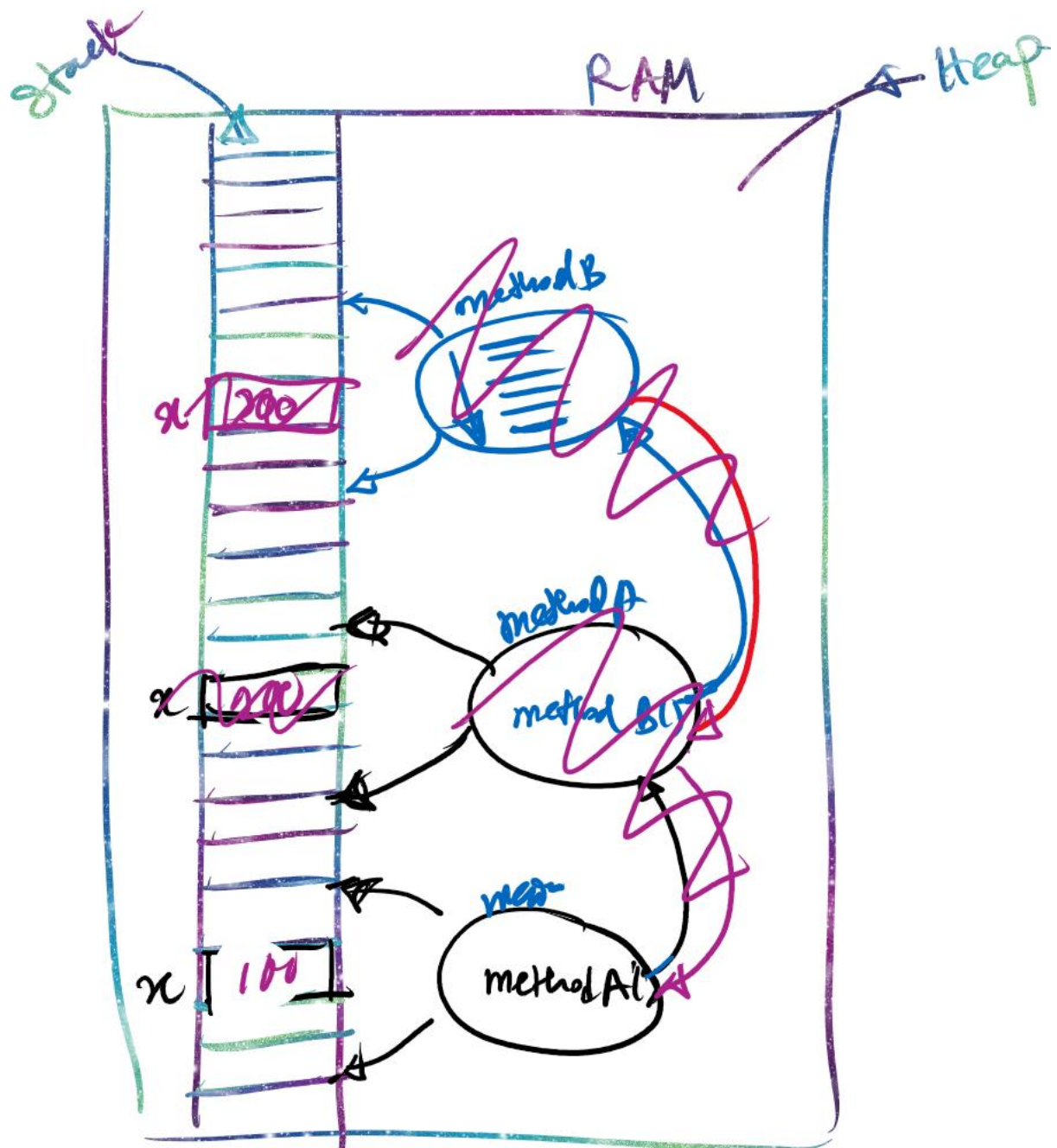
```

Start main();
 Start methodA();
 Start methodB();
 End methodB();
 End methodA();
 End main();

Q298



Start main()
 Start methodA()
 Start methodB()
 End methodB()
 End methodA()
 End main()



////////////////////////////////////

=====

////////////////////////////////////

=====

////////////////////////////////////

=====

////////////////////////////////////

=====

////////////////////////////////////

=====

////////////////////////////////////

Q305 Case V

=====

```
class Example{
    public static void methodA(int x) {

    }

    public static void main(String args[]){

        byte b = 100;
        short s = 100;
        int i = 100;
        long l = 100;
        float f = 100.0f;
        double d = 100.0;
        char ch = 'A';
        boolean bool = false;

        methodA(b);
        methodA(s);
        methodA(i);
        methodA(l); // Illegal
        methodA(f); // Illegal
        methodA(d); // Illegal
        methodA(ch);
        methodA(bool); // Illegal
    }
}
```

//

Q306 Case VI

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```
class Example{
    public static void methodA(int x) {

    }

    public static void main(String args[]){

        byte b = 100;
        short s = 100;
        int i = 100;
        long l = 100;
        float f = 100.0f;
        double d = 100.0;
```


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}

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```
class Example{
    public static int methodA(int x) {
        if( x > 100) {
            return 0;
        }
        return 1;
    }
}
```


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}

```
A.methodB();
```

}

}

////////////////////////////////////

Q330

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Q331

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Q332

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Q333

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Q334

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Q335

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Q336

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Q337

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Q338

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Q339

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Q340